Comparison of Multiple Machine Learning Approaches for Sex-Specific Survival Modelling in Meningioma

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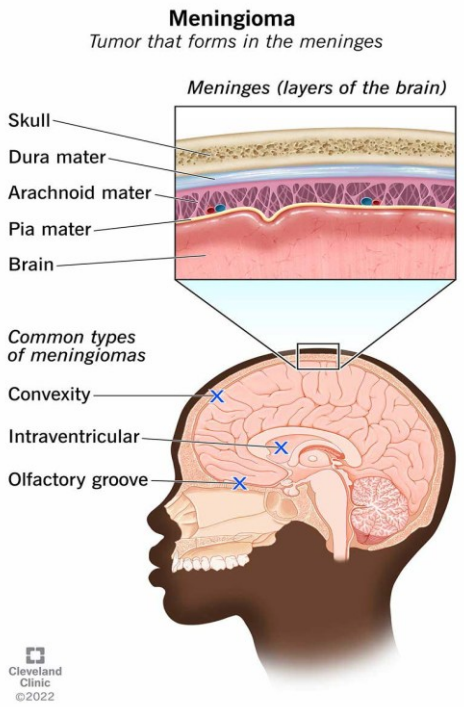
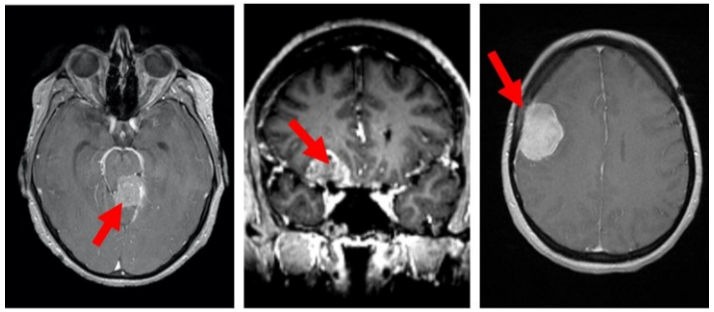
Clinical and Epidemiological Overview of Meningioma

- Meningiomas are among the most common intracranial tumors, with an estimated incidence of **8.83 cases per 100,000** persons per year
- Meningioma accounts for **36.3%** of primary intracranial tumors of the central nervous system.
- Meningiomas account for **40% of all brain neoplasms** globally.
- The majority are **intracranial (81.2%)**, while **4.2% are spinal**.
- It arises from the meninges, the protective membrane that covers the brain and spinal cord.
- Meningioma incidence increases with age.
- The median age at diagnosis is 66 years.
- For patients aged 40+, the incidence rate is 18.69 per 100,000.
- In those aged 0–19, the rate drops to 0.16 per 100,000.
- Among CNS tumors:
    - Meningiomas make up 43.6% of people aged 40+.
    - They account for 15.6% of those aged 15–39.
    - Only 1.7% in ages 0–14.



Source: my.clevelandclinic.org



Source: neurosurgery.com

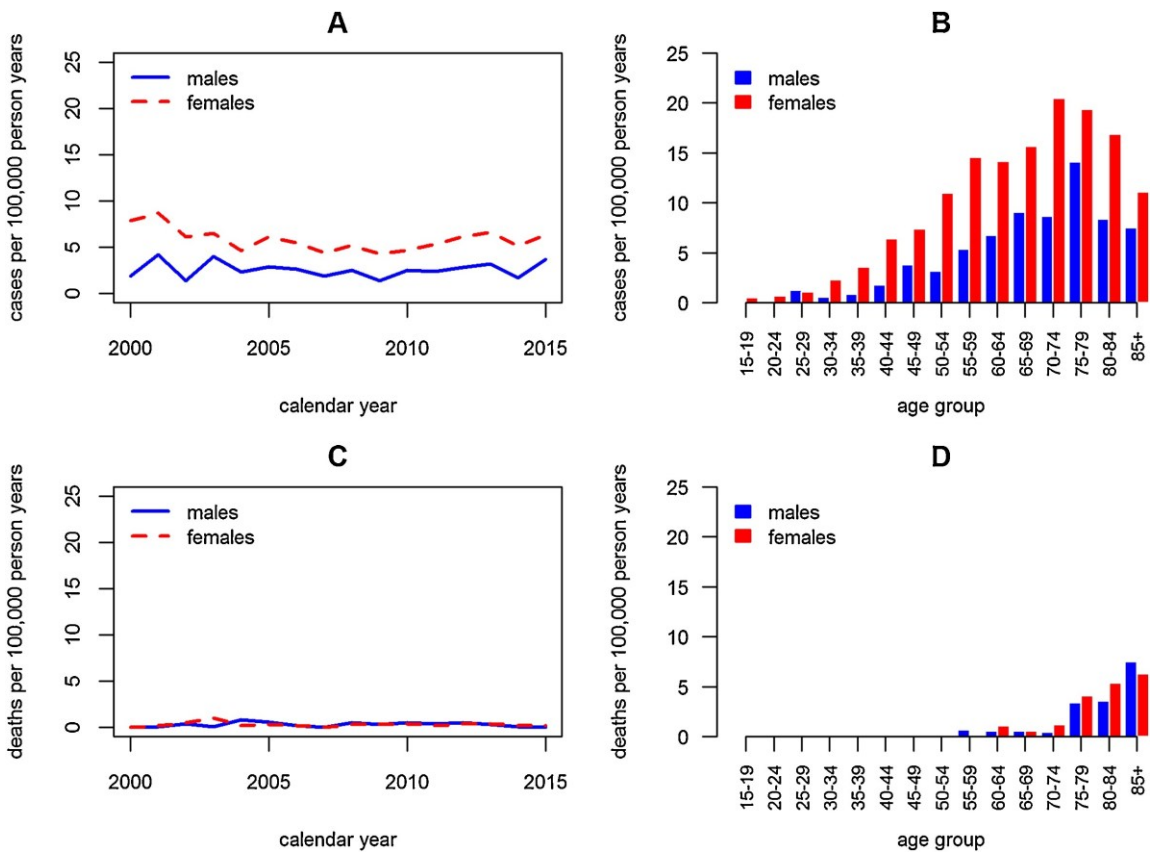
Updated WHO 2021 Diagnostic Meningioma Classification.

	Grade 1	Grade 2	Grade 3
Prevalence	~85-95%	~5-10%	~1-5%
Gender	Female > Male	Female > Male	Female < Male
Diagnostic Criteria	Mitoses < 4/10 hpf	Mitoses 4 19/10 hpf OR 3/5 of the following: Necrosis High nuclear/cytoplastic ratio Prominent nucleoli Architectural sheeting Hypercellularity OR Clear cell/chordoid histology - Brain Invasion	Mitoses ≥ 20/10 hpf OR Frank anaplasia OR Papillary/rhabdoid histology
Histologic Subtypes	Meningothelial Fibrous (Fibroblastic) Transitional (Mixed) Psammomatous Angiomatous Microcystic Secretory Lymphoplasmacyte-rich Metaplastic	Atypical Chordoid Clear Cell	Anaplastic Papillary Rhabdoid
Clinical Outcomes			
Overall Survival	80-90%	50-79%	14-34%
Progression-Free Survival	75-90%	23-78%	0%

Source: 10.3389/fonc.2023.1137849

Sex-Based Disparities in Meningioma Incidence and Progression

- Benign meningiomas are more common in females (**IRR 2.33**).
- Women are **2 to 2.5 times** more likely to develop meningiomas than men.
- The disparity is **most pronounced during reproductive years**.
- WHO grade 1 meningiomas are **2.3 times** more common in women, with the greatest risk between ages 35 and 54.
- Meningioma cells often express estrogen and progesterone receptors.
- Hormonal changes such as menstruation, pregnancy, and hormone replacement therapy are linked to increased tumor growth in women.
- While the overall incidence is higher in females, the risk of recurrence and aggressive tumor behavior presents a contrasting pattern.
- **Males** have a **higher risk of early recurrence** after surgery, especially in **WHO Grade 1** meningiomas.
- Males have significantly shorter progression-free survival compared to females.
- Sex-specific genetic differences may influence tumor behavior and prognosis in meningioma patients.

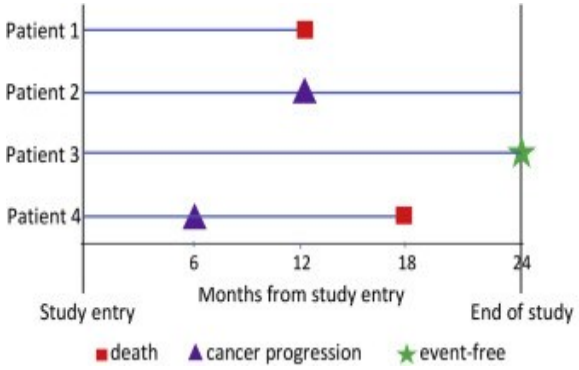


Source: *Incidence, mortality and outcome of meningiomas: A population-based study from Germany*

Machine Learning in Recurrence-Free Survival Analysis

What is Survival Analysis?

- Survival analysis is a statistical approach used to analyze and model the **time until an event of interest occurs**.
- Accounts for both completed and censored observations.
- Help link molecular data to patient outcomes.
- Identify prognostic biomarkers.
- Further, we can determine which patients are at higher risk for recurrence or death.
- Helps clinicians make informed decisions backed by statistical evidence.

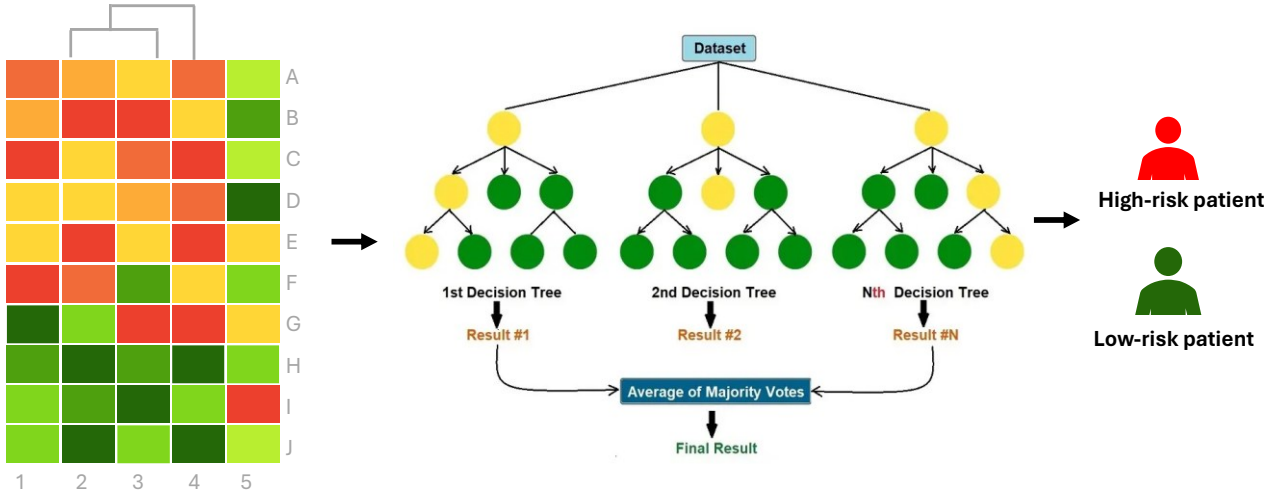


Patient	Observed data as composite progression-free survival endpoint (indicator: 1 = progression or death, 0 = censored)		Observed data as competing risks (indicator: 1 = progression, 2 = death, 0 = censored)	
	Time (months)	Indicator	Time (months)	Indicator
1	12	1	12	2
2	12	1	12	1
3	24	0	24	0
4	6	1	6	1

Source: [Time-to-event data: an overview and analysis considerations]

Role of Machine Learning in Recurrence-Free Survival Analysis

- Patient RFS data is usually analyzed using traditional survival models.
- These statistical models may not often capture the complexities inherent in medical data.
- Machine learning techniques can capture the subtle relevance and non-linear relationships among variables.
- Achieve better model performance than conventional statistical methods.



Objectives

Sex-Stratified Differential Expression Analysis
Perform a sex-stratified differential expression analysis between recurrent and non-recurrent meningioma samples using publicly available RNA-seq datasets.

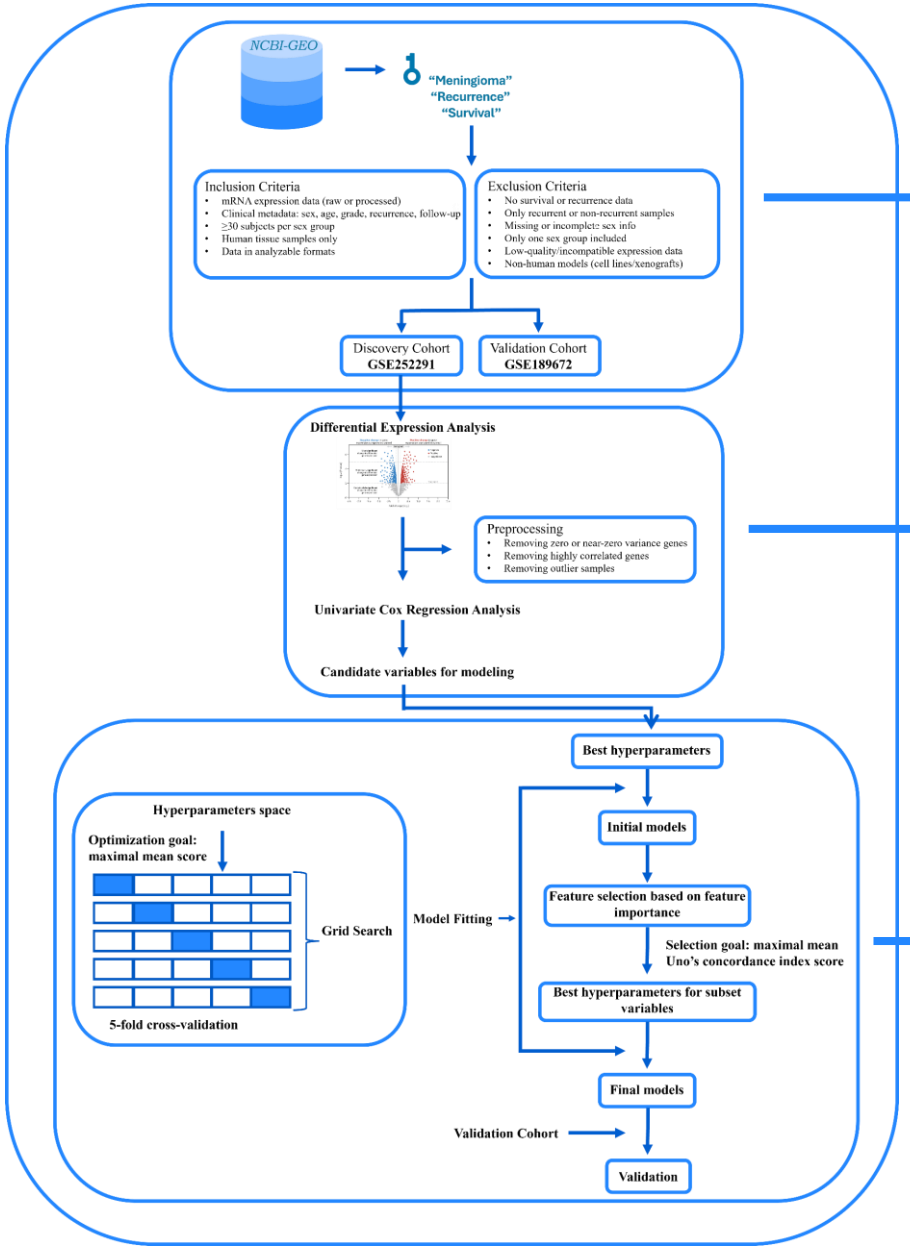
Dimensionality Reduction
Dimensionality reduction using Univariate Cox Proportional Hazards and identifying genes significantly associated with recurrence-free survival in a sex-specific manner.

Machine Learning Based Survival Modelling
To evaluate and compare the performance of multiple machine learning–based survival models in predicting recurrence-free survival in female and male cohorts independently.

External Validation
Apply the final, tuned models to an independent external validation cohort and assess their generalizability using the same set of genes.

Cross-Gender Validation
Conduct cross-sex application of trained models (applying female-trained models to male patients and vice versa) to evaluate the specificity and performance decay of sex-independent modelling.

Study Design



Data Acquisition

Feature Selection

Model Development and Validation

Strategy for Data Collection

"Meningioma"
"Recurrence"
"Survival"

Inclusion Criteria

- mRNA expression data (raw or processed)
- Clinical metadata: sex, age, grade, recurrence, follow-up
- ≥30 subjects per sex group
- Human tissue samples only
- Data in analyzable formats

Exclusion Criteria

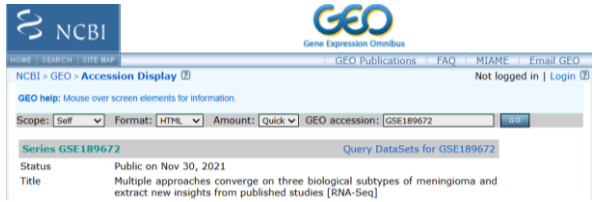
- No survival or recurrence data
- Only recurrent or non-recurrent samples
- Missing or incomplete sex info
- Only one sex group included
- Low-quality/incompatible expression data
- Non-human models (cell lines/xenografts)

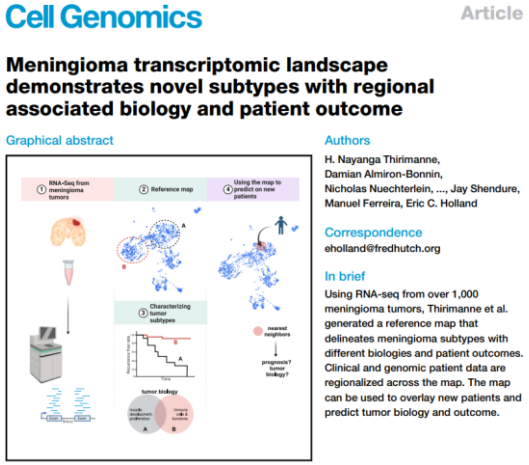
Discovery Cohort
GSE252291
n=255

Validation Cohort
GSE189672
n=110

Discovery Cohort

Validation Cohort





Cell Genomics — Article

Meningioma transcriptomic landscape demonstrates novel subtypes with regional associated biology and patient outcome

Graphical abstract

Authors
H. Nayanga Thirimanne, Damian Almiron-Bonnin, Nicholas Nuechterlein, ..., Jay Shendure, Manuel Ferreira, Eric C. Holland

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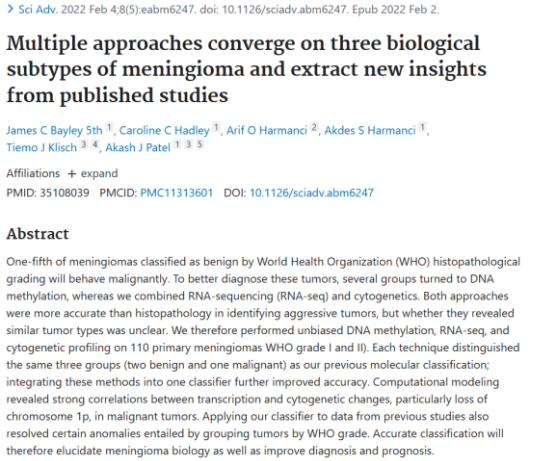
In brief
Using RNA-seq from over 1,000 meningioma tumors, Thirimanne et al. generated a reference map that delineates meningioma subtypes with different biologies and patient outcomes. Clinical and genomic patient data are regionalized across the map. The map can be used to overlay new patients and predict tumor biology and outcome.



Sci Adv. 2022 Feb 4;8(5):eabm6247. doi: 10.1126/sciadv.abm6247. Epub 2022 Feb 2.

Multiple approaches converge on three biological subtypes of meningioma and extract new insights from published studies

James C Bayley 5th [1], Caroline C Hadley [1], Arif O Harmanci [2], Akdes S Harmanci [1], Tiemo J Klisch [3 4], Akash J Patel [1 3 5]

Affiliations + expand

PMID: 35108039 PMCID: PMC11313601 DOI: 10.1126/sciadv.abm6247

Abstract

One-fifth of meningiomas classified as benign by World Health Organization (WHO) histopathological grading will behave malignantly. To better diagnose these tumors, several groups turned to DNA methylation, whereas we combined RNA-sequencing (RNA-seq) and cytogenetics. Both approaches were more accurate than histopathology in identifying aggressive tumors, but whether they revealed similar tumor types was unclear. We therefore performed unbiased DNA methylation, RNA-seq, and cytogenetic profiling on 110 primary meningiomas WHO grade I and II). Each technique distinguished the same three groups (two benign and one malignant) as our previous molecular classification; integrating these methods into one classifier further improved accuracy. Computational modeling revealed strong correlations between transcription and cytogenetic changes, particularly loss of chromosome 1p, in malignant tumors. Applying our classifier to data from previous studies also resolved certain anomalies entailed by grouping tumors by WHO grade. Accurate classification will therefore elucidate meningioma biology as well as improve diagnosis and prognosis.

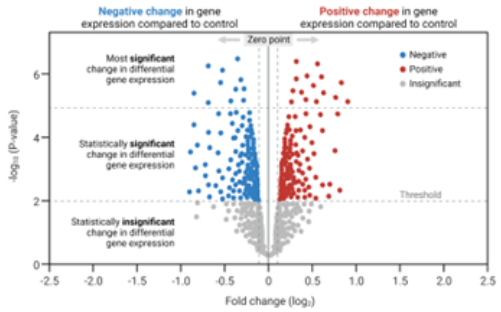
Strategy for Data Collection

Characteristic	Discovery Cohort (GSE252291)		Validation Cohort (GSE189672)*	
	Females	**Males**	**Females**	**Males**
Total Samples (n)	156	69	67	43
Median Age (years)	57 (IQR: 43.75–66)	54 (IQR: 41–64)	59 (IQR: 47–67)	60 (IQR: 51.5–71.0)
WHO Grade**				
– Grade I	54.49%	40.58%	89.55%	69.77%
– Grade II	41.67%	49.28%	10.45%	30.23%
– Grade III	3.21%	8.70%	-	-
Recurrence Status				
– Recurrent	24.36%	47.83%	14.93%	20.93%
– Non-recurrent	75.64%	52.17%	85.07%	79.07%
Median Time to Recurrence or Censoring (months)***	315 (IQR: 64–315)	96 (IQR: 19–315)	24.3 (IQR: 13.18–45.43)	22.23 (IQR: 5.22–41.97)

*No WHO Grade III tumors were present in the validation cohort.
**WHO grade information was unavailable for a subset of samples.
***Time to recurrence or censoring was unavailable for all patients due to missing follow-up.

Feature Selection for ML Models

Differential Expression Analysis



Preprocessing

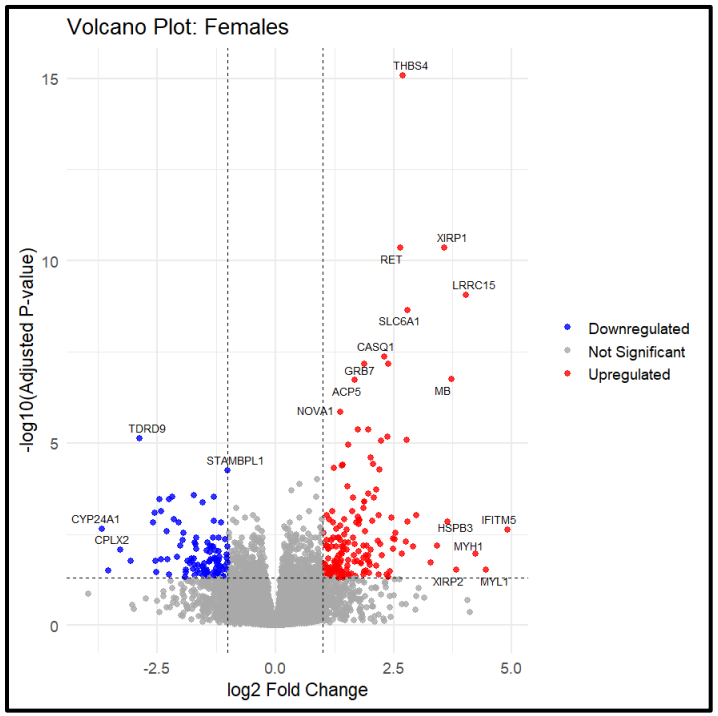
- Removing zero or near-zero variance genes
- Removing highly correlated genes
- Removing outlier samples

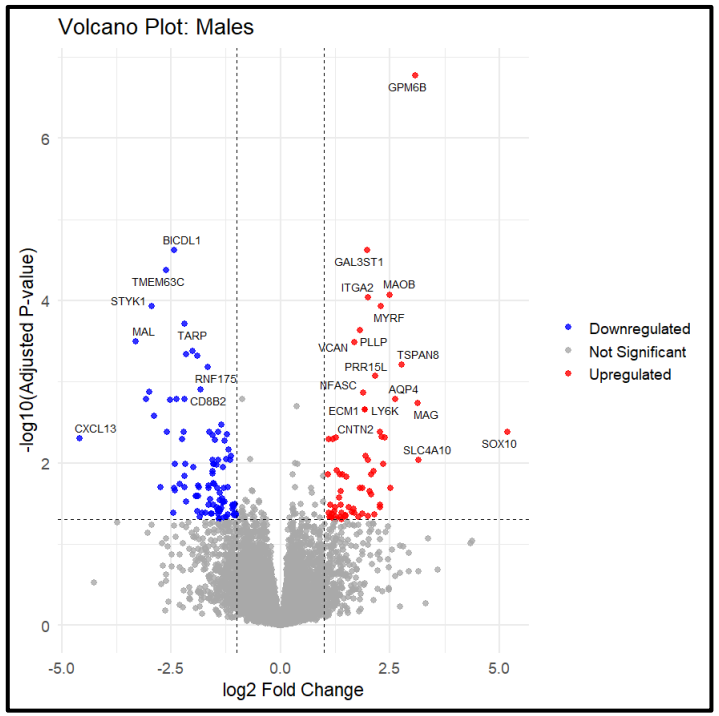
Univariate Cox Regression Analysis

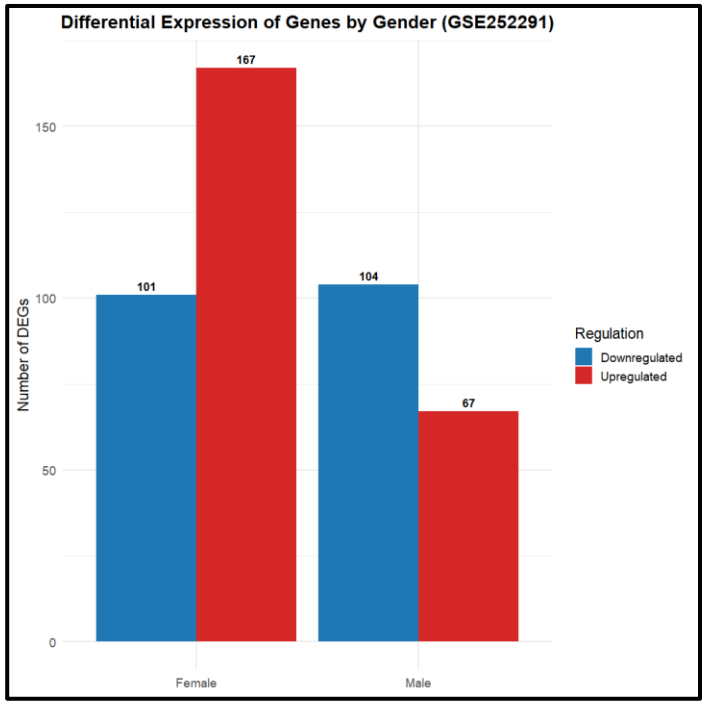
Candidate variables for modeling

Differential Expression Analysis

- Performed Differential Expression Analysis in the sex-stratified Recurrent vs Non-Recurrent samples.
- **DESeq2, R**
- Normalization: **Median of Ratios**
- Noise Removal: **HTSFilter**
- Gene Selection: **p-adj < 0.05, log2 fold change > 1**
- The normalized values were transformed using the **varianceStabilizingTransformation()** for further preprocessing.







Genes	Females	Males
KLRC2	1.75	-2.17
KRT81	-1.31	-1.23
LRRTM3	1.54	2.07
MGARP	-1.24	-2.38
SCN9A	1.79	1.87
TRIM63	1.52	2.14

Univariate Cox Regression Analysis

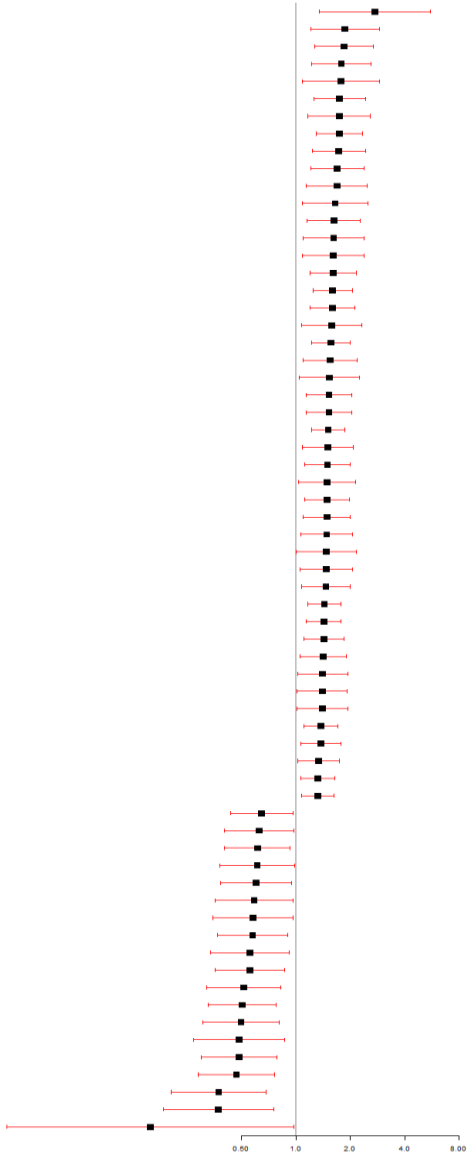
- Univariate Cox Regression analysis examines the relationship between a single predictor variable and the hazard of an event over time, without considering other factors.
- To identify individual genes significantly associated with recurrence-free survival.
- Performed using the **coxph()** function from the survival package (version 2.34) in R.
- Input: Expression levels of each gene and corresponding time to event data.
- Output: Hazard ratios (HR), confidence intervals, and p-value.
- HR > 1 indicates **risk factor**, HR < 1 indicates **protective effect**.
- Genes with p-value < 0.05 were considered statistically significant.

```r
library(survival)

# Load data
data <- read.csv("File.csv", row.names = 1)

# Extract Time and Event
Time <- data$Time
Event <- data$Event

# Subset expression data (from 4th column onwards)
expression_data <- data[, -(1:2)]  # removes Time and Event

# Perform univariate Cox regression for each lncRNA
results <- data.frame(genes = colnames(expression_data),
                      beta = NA,
                      HR = NA,
                      p_value = NA,
                      lower_CI = NA,
                      upper_CI = NA)

for (i in 1:ncol(expression_data)) {
  genes <- colnames(expression_data)[i]
  cox_model <- coxph(Surv(Time, Event) ~ expression_data[, i])
  summary_model <- summary(cox_model)
  results[i, "beta"] <- summary_model$coefficients[1, "coef"]
  results[i, "HR"] <- summary_model$coefficients[1, "exp(coef)"]
  results[i, "p_value"] <- summary_model$coefficients[1, "Pr(>|z|)"]
  results[i, "lower_CI"] <- summary_model$conf.int[1, "lower .95"]
  results[i, "upper_CI"] <- summary_model$conf.int[1, "upper .95"]
}
results$FDR <- p.adjust(results$p_value, method = "BH")

# Save results
write.csv(results, "Results.csv", row.names = FALSE)
```

Forest plots of significant genes associated with recurrence-free survival identified by Univariate Cox Regression Analysis

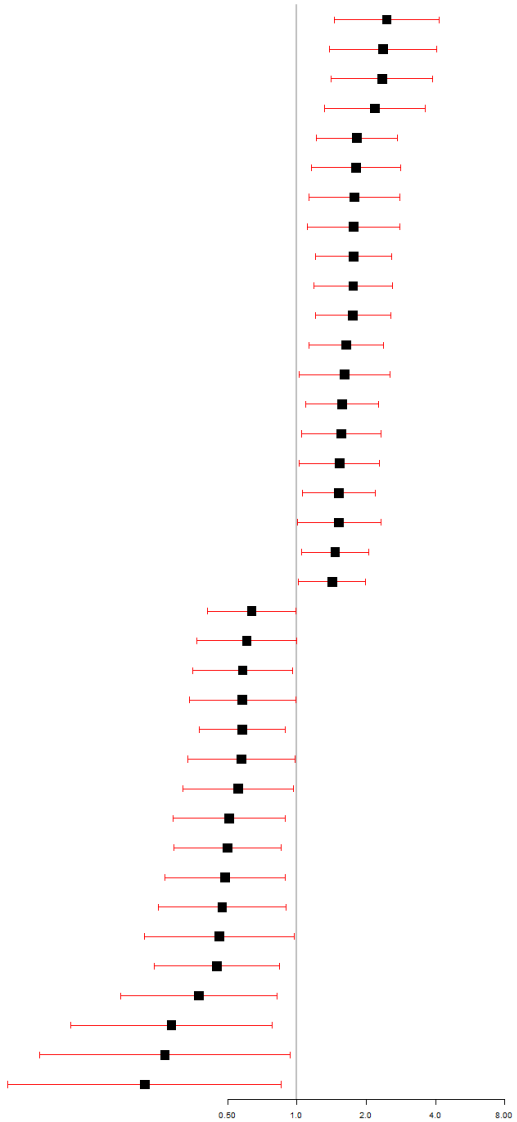
Forest Plot of Significant Genes in Females

Gene	HR (95% CI)	p-value
SLC36A2	2.73 (1.34-5.56)	0.0055
CHAD	1.87 (1.21-2.91)	0.0052
CDH15	1.85 (1.28-2.68)	0.0012
HSPB3	1.79 (1.22-2.61)	0.0029
CFAP58	1.78 (1.09-2.91)	0.022
SPINK2	1.75 (1.26-2.43)	0.00095
NXPH3	1.74 (1.17-2.59)	0.0063
C6orf118	1.74 (1.29-2.33)	0.00024
RET	1.73 (1.23-2.43)	0.0017
L1CAM	1.7 (1.21-2.39)	0.0025
ERVMER34.1	1.69 (1.15-2.48)	0.0079
TRPM6	1.65 (1.08-2.51)	0.02
MT1A	1.62 (1.16-2.27)	0.0051
PITX1	1.62 (1.1-2.38)	0.014
LRRC15	1.61 (1.09-2.39)	0.018
SCN9A	1.61 (1.2-2.16)	0.0015
TRIM54	1.6 (1.24-2.06)	0.00024
XIRP1	1.59 (1.2-2.11)	0.0012
INSC	1.58 (1.08-2.31)	0.019
APOBEC2	1.56 (1.22-2)	0.00045
USP2	1.55 (1.1-2.18)	0.012
POSTN	1.53 (1.04-2.24)	0.03
DNER	1.52 (1.14-2.03)	0.004
JSRP1	1.52 (1.14-2.04)	0.0043
RNF17	1.51 (1.23-1.86)	1e-04
SMTNL2	1.51 (1.09-2.09)	0.014
CASQ1	1.5 (1.12-1.99)	0.0059
MRGPRX2	1.49 (1.04-2.14)	0.031
THBS4	1.49 (1.11-1.99)	0.007
RAPSN	1.48 (1.1-2)	0.01
HABP2	1.48 (1.06-2.06)	0.02
STAP1	1.47 (1.01-2.16)	0.046
IGSF5	1.47 (1.05-2.06)	0.024
SLC22A8	1.47 (1.08-2)	0.015
IGFN1	1.44 (1.16-1.77)	0.00072
DES	1.43 (1.15-1.78)	0.0016
AGBL1	1.43 (1.11-1.84)	0.0062
DOK7	1.41 (1.05-1.9)	0.021
MARVELD3	1.41 (1.02-1.94)	0.037
RAB44	1.4 (1.02-1.93)	0.038
CADM3	1.4 (1.01-1.94)	0.044
SYNPO2L	1.38 (1.11-1.71)	0.0037
FOXD3	1.38 (1.07-1.78)	0.014
LBX1	1.34 (1.03-1.74)	0.03
MB	1.33 (1.07-1.64)	0.01
CACNA1S	1.32 (1.08-1.62)	0.0078
CLEC5A	0.65 (0.43-0.96)	0.032
CLIC5	0.63 (0.4-0.98)	0.039
XCL1	0.61 (0.4-0.93)	0.021
TRIM67	0.61 (0.38-0.98)	0.042
RNF43	0.6 (0.38-0.94)	0.027
RNASE3	0.59 (0.36-0.97)	0.036
C12orf42	0.58 (0.35-0.96)	0.035
HSPA12A	0.58 (0.37-0.9)	0.015
LHCGR	0.55 (0.34-0.91)	0.021
DACT2	0.55 (0.36-0.86)	0.0091
PRAM1	0.51 (0.32-0.82)	0.0054
STAMBPL1	0.5 (0.33-0.78)	0.002
MYCN	0.5 (0.3-0.81)	0.0053
INA	0.49 (0.27-0.87)	0.015
CD244	0.48 (0.3-0.78)	0.0032
NLRP12	0.47 (0.29-0.76)	0.0023
CCR2	0.37 (0.2-0.69)	0.0015
ADGRE3	0.37 (0.19-0.75)	0.0059
CPLX2	0.16 (0.03-0.98)	0.047

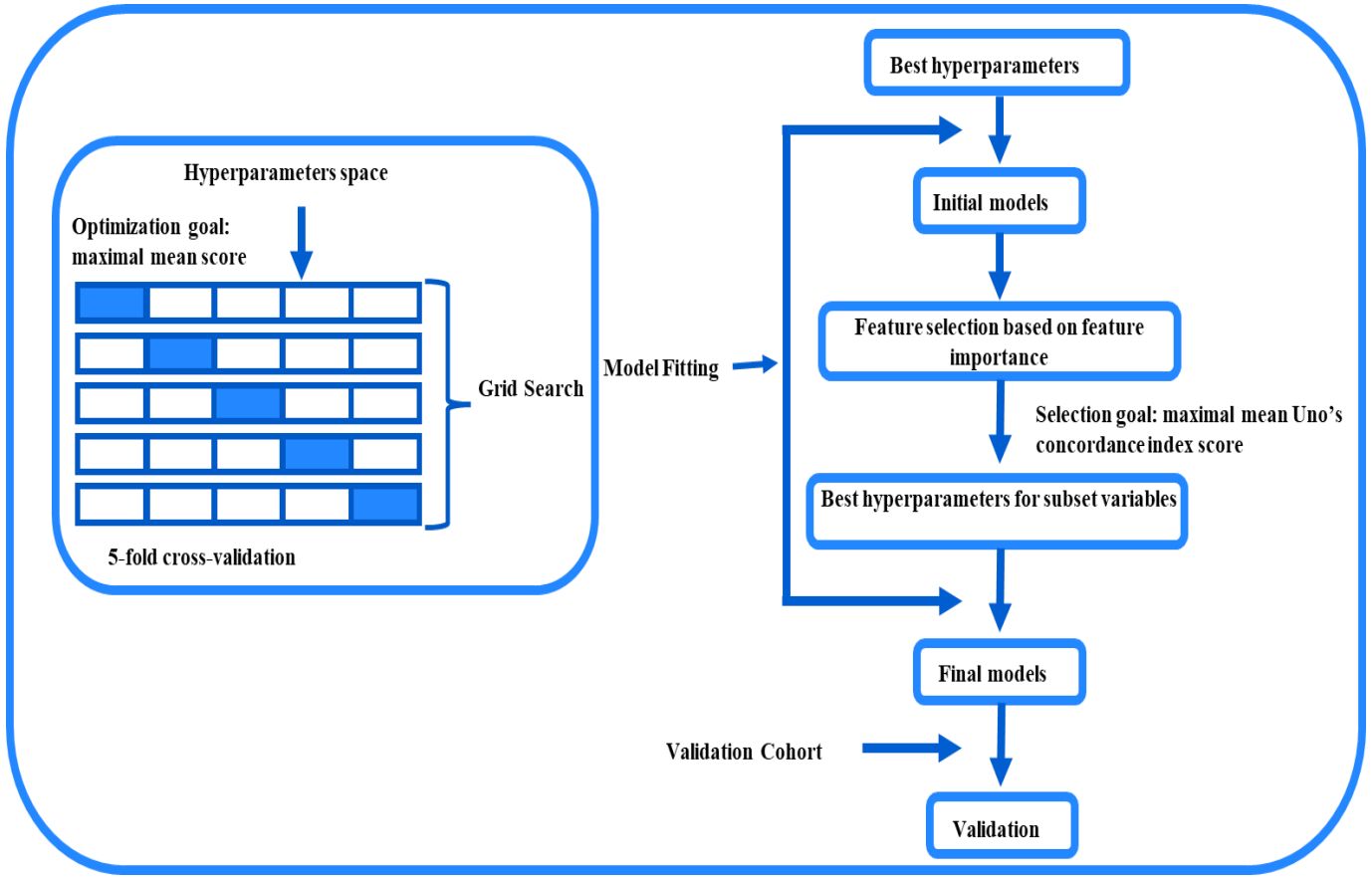


Forest Plot of Significant Genes in Males

Gene	HR (95% CI)	p-value
VCAN	2.46 (1.45-4.16)	8e-04
AQP4	2.37 (1.38-4.05)	0.0017
GAL3ST1	2.34 (1.41-3.9)	0.001
CDKN2B	2.19 (1.32-3.62)	0.0024
GPM6B	1.82 (1.21-2.73)	0.0038
PGR	1.81 (1.16-2.82)	0.0089
CRLF1	1.77 (1.13-2.8)	0.013
LRRTM3	1.76 (1.11-2.79)	0.016
TFPI2	1.76 (1.2-2.57)	0.0036
USH2A	1.76 (1.19-2.6)	0.0047
CHST9	1.75 (1.21-2.55)	0.0033
ITGA2	1.64 (1.13-2.39)	0.0096
NRCAM	1.61 (1.02-2.55)	0.041
NOX4	1.57 (1.09-2.27)	0.015
BAALC	1.56 (1.05-2.32)	0.028
SHC4	1.53 (1.02-2.29)	0.038
C2CD4A	1.52 (1.06-2.19)	0.024
TRIM7	1.52 (1-2.31)	0.049
SULT1B1	1.47 (1.05-2.05)	0.027
ANKRD18B	1.42 (1.02-1.98)	0.039
TCEAL5	0.64 (0.41-0.99)	0.046
TGM2	0.6 (0.37-1)	0.049
AGTR1	0.58 (0.35-0.95)	0.032
JPH4	0.58 (0.34-0.99)	0.046
TUBA4A	0.58 (0.38-0.89)	0.012
JAK3	0.57 (0.34-0.98)	0.042
CPNE5	0.56 (0.32-0.97)	0.038
CTTNBP2	0.51 (0.29-0.89)	0.018
GLB1L3	0.5 (0.29-0.85)	0.011
CRYAB	0.49 (0.27-0.89)	0.018
ANGPTL1	0.47 (0.25-0.89)	0.021
FAIM2	0.46 (0.22-0.97)	0.041
HECW1	0.45 (0.24-0.84)	0.012
TMEM63C	0.37 (0.17-0.82)	0.014
CD8B	0.29 (0.1-0.78)	0.015
ECE2	0.27 (0.08-0.93)	0.039
MGARP	0.22 (0.06-0.85)	0.029



Machine Learning Based Model Development and Validation for Recurrence-Free Survival




```
from tqdm import tqdm
from datetime import datetime, date
import joblib
```

In [2]:

```
# Load data
data = pd.read_csv("C:/Users/SAKSHI/OneDrive - Jamia Millia Islamia (A Central University)/Disseration/ML_input/Females_vst_

# Features and target
X = data.drop(columns=["samples", "time", "event"])
y = Surv.from_dataframe(event="event", time="time", data=data)

# Train/test split (stratified)
# X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=123, stratify=data["event"])

def train_test_generator(X, y, stratified=None, n_splits=2):
    if stratified is not None:
        return StratifiedKFold(n_splits=n_splits).split(X, y[stratified])
    else:
        return KFold(n_splits=n_splits).split(X)


for stratified in [None, "event"]:
    print(f"stratified={stratified}")
    for i, (train_index, test_index) in enumerate(train_test_generator(X, y, stratified=stratified)):
        print(f'split {i+1}: {y[train_index]["event"].sum()} + {y[test_index]["event"].sum()} = {y["event"].sum()}')
```

```
stratified=None
split 1: 12 + 14 = 26
split 2: 14 + 12 = 26
stratified=event
split 1: 13 + 13 = 26
split 2: 13 + 13 = 26
```

In [6]:

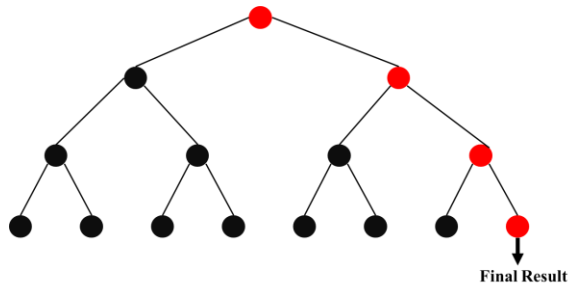
```
# Define parameter grid
# Define parameter grid
parameters = {
    'n_estimators': [10, 20, 30],
    'min_samples_split': [2, 5, 7, 10],
```

Machine Learning Based Model Development and Validation for Recurrence-Free Survival

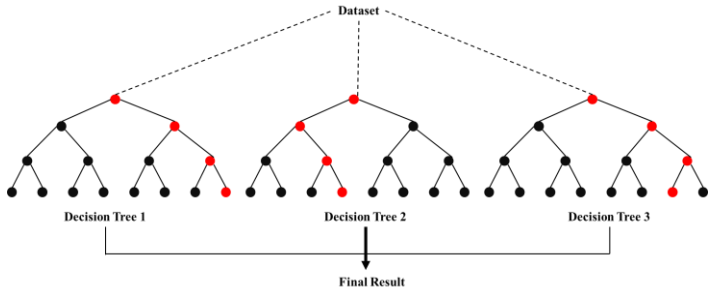
1. Tree-Based Models

- **Random Survival Forest (RSF)**
- **Survival Tree**

These models were used to capture non-linear relationships and complex feature interactions. RSF, in particular, is an ensemble method that can handle high-dimensional data, account for interactions automatically, and is robust to overfitting. Survival Trees provide interpretability and serve as a simpler tree-based counterpart for comparison.



Survival Tree

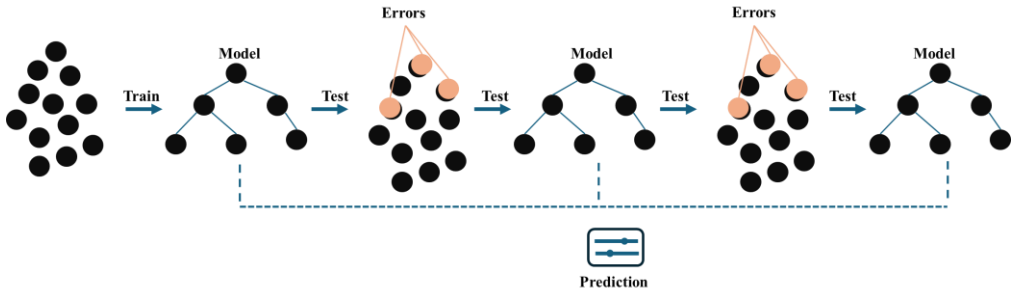


Random Survival Forest

2. Boosting Models

- **Gradient Boosting Survival Analysis (GBS)**
- **Component-wise Gradient Boosting Survival Analysis (CWGBS)**

Boosting algorithms build predictive models in an additive, stage-wise fashion by minimizing loss functions. These were employed to iteratively improve weak learners, capturing subtle signal patterns that may not be apparent to simpler models. The inclusion of both GBS and CWGBS allowed comparison between full-feature and component-wise boosting strategies, enhancing flexibility and interpretability.



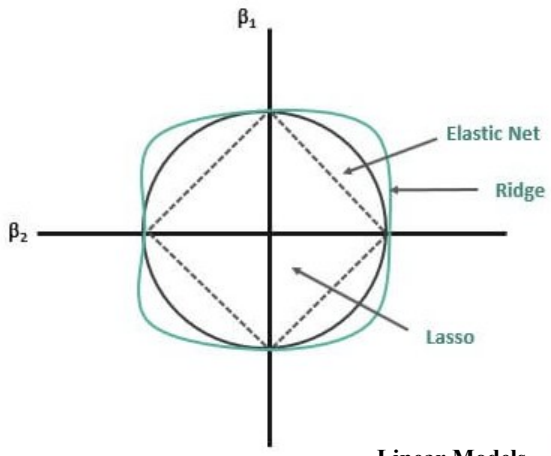
Gradient Boosting Model

Machine Learning Based Model Development and Validation for Recurrence-Free Survival

3. Linear Models (Regularized Cox)

- **CoxNet (LASSO/ElasticNet Penalized Cox Regression)**

These models extend the classical Cox proportional hazards model by incorporating L1 (LASSO) and combined (ElasticNet) regularization. They were chosen to interpret the linear relationships in the data.



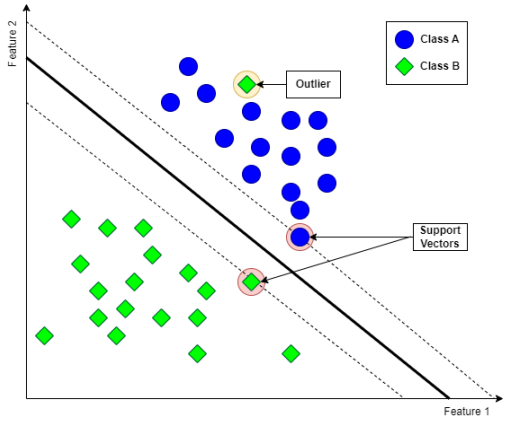
Linear Models

Source: wallstreetmojo.com

4. Support Vector Machine Models

- **Fast Kernel Survival SVM (FKSSVM)**

SVM-based survival models were used to explore margin-based optimization approaches for censored data. FKSSVM uses kernel transformations to model non-linear relationships. This model provides an alternative to tree and regression-based approaches, complementing the modeling diversity.

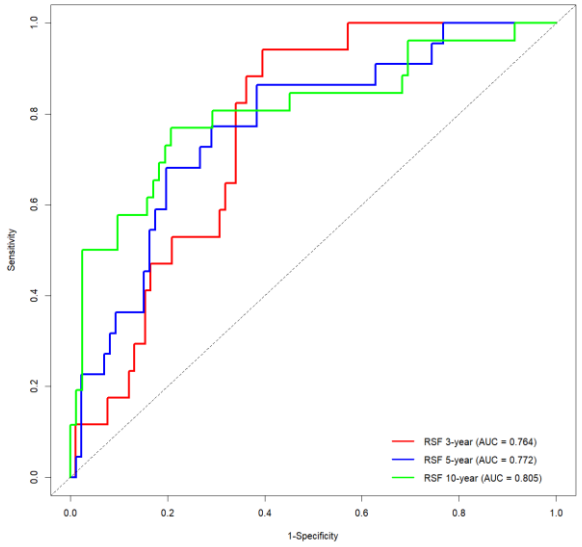


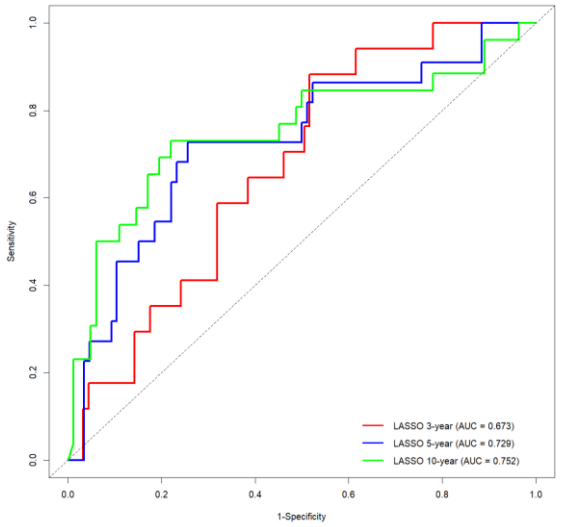
Support Vector Machine

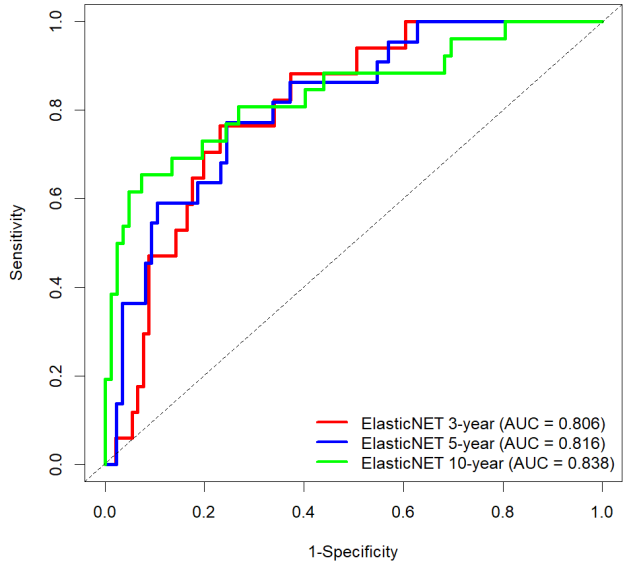
Source: towardsdatascience.com

Model Performance in Female Cohort (Internal Validation)

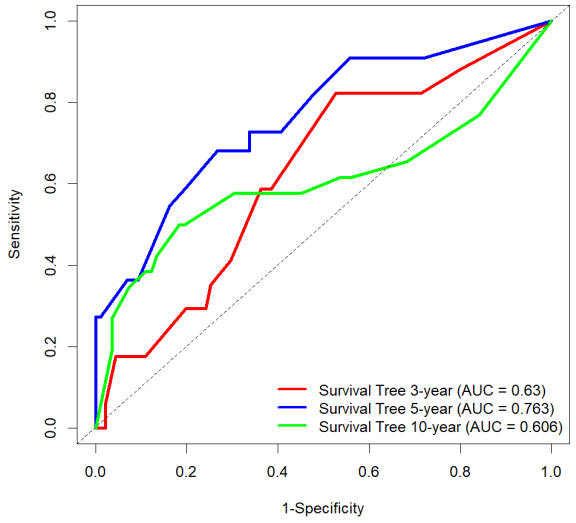
3-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.847	0.093	0.76
Survival Tree	0.625	0.102	0.63
Component-wise GBM	0.678	0.102	0.33
GBM	0.745	0.100	0.43
ElasticNET	0.858	0.104	0.81
LASSO	0.771	0.118	0.67
Fast Kernel SVM	0.736	-	0.52
Coxph	0.5	0.276	0.5
5-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.770	0.109	0.77
Survival Tree	0.745	0.102	0.76
Component Wise GBM	0.587	0.124	0.48
GBM	0.731	0.122	0.61
ElasticNET	0.849	0.119	0.82
LASSO	0.781	0.131	0.73
Fast Kernel SVM	0.708	-	0.65
Coxph	0.5	0.295	0.5
10-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.819	0.128	0.81
Survival Tree	0.608	0.175	0.61
Component Wise GBM	0.636	0.150	0.62
GBM	0.751	0.147	0.71
ElasticNET	0.864	0.119	0.84
LASSO	0.785	0.140	0.75
Fast Kernel SVM	0.755	-	0.76
Coxph	0.5	0.336	0.5

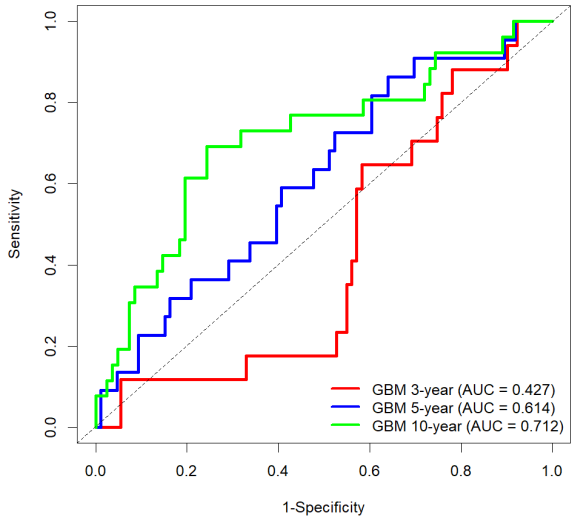


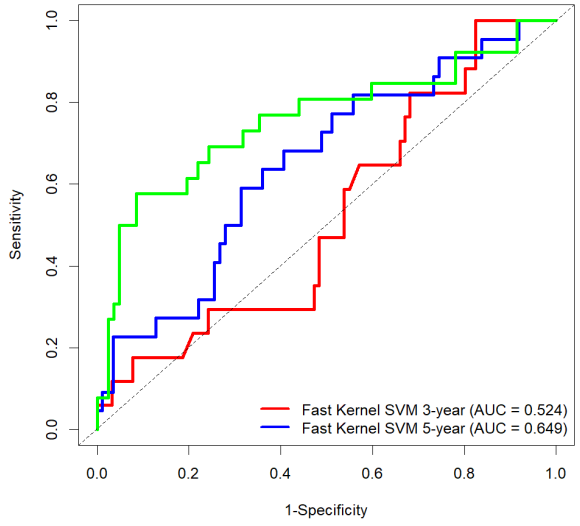


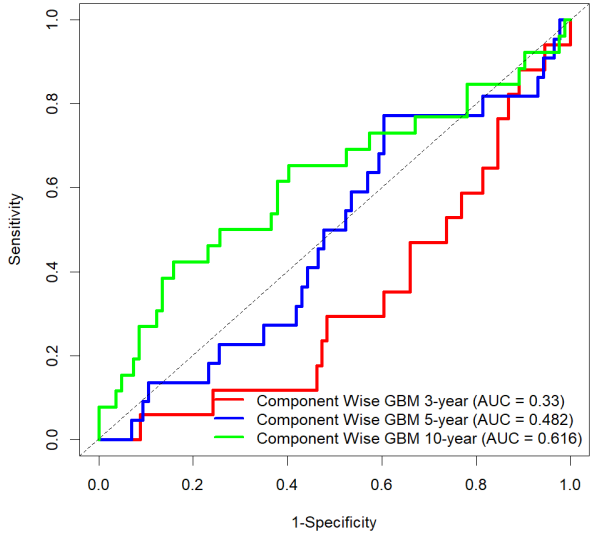


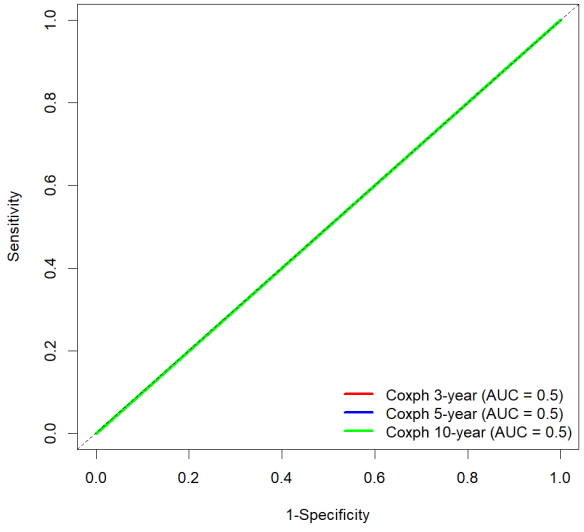
Model Performance in Female Cohort (Internal Validation)





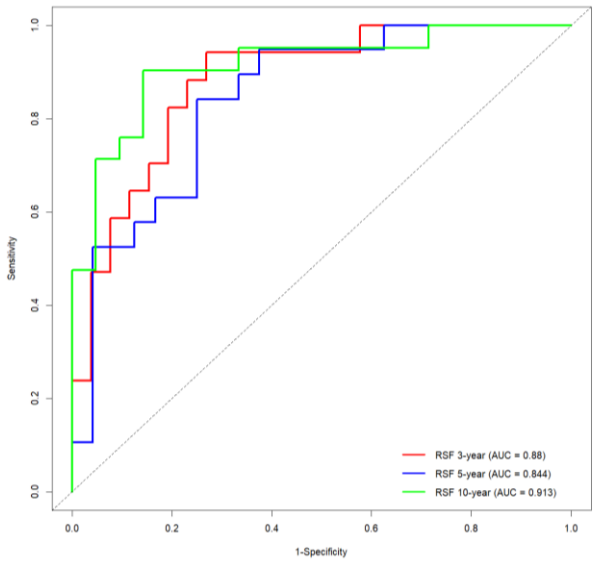


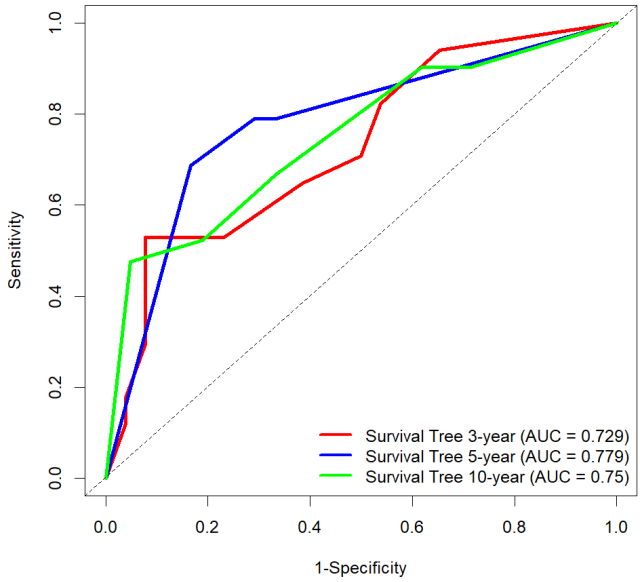


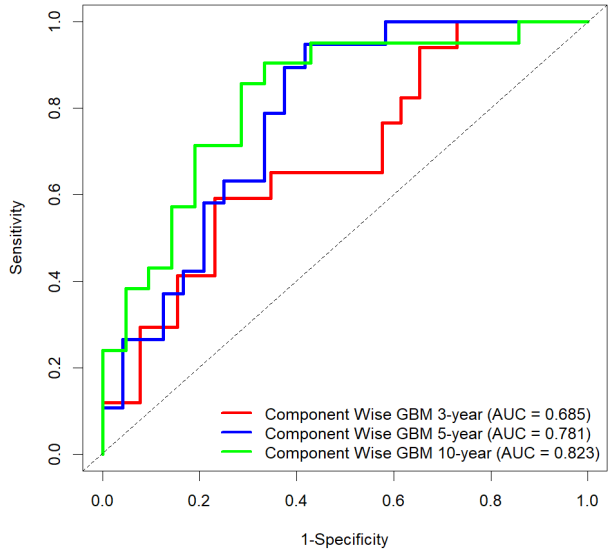


Model Performance in Male Cohort (Internal Validation)

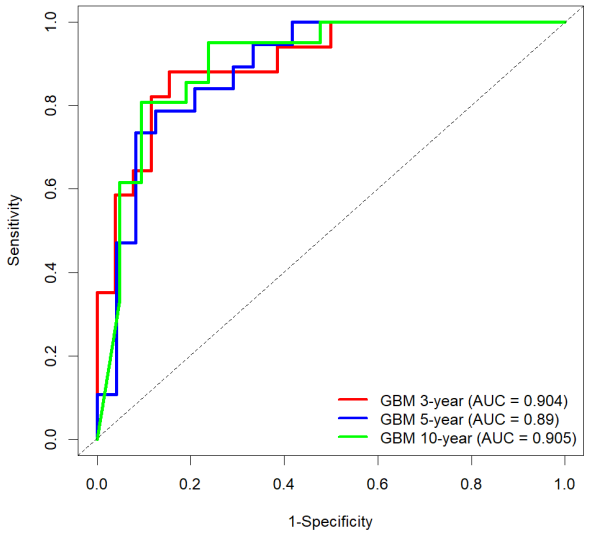
3-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.891	0.173	0.88
Survival Tree	0.756	0.211	0.73
Component Wise GBM	0.842	0.193	0.69
GBM	0.884	0.175	0.90
ElasticNET	0.881	0.201	0.87
LASSO	0.826	0.191	0.86
Fast Kernel SVM	0.893	-	0.90
Coxph	0.5	0.258	0.5
5-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.848	0.163	0.84
Survival Tree	0.704	0.246	0.78
Component Wise GBM	0.779	0.200	0.78
GBM	0.883	0.161	0.89
ElasticNET	0.842	0.212	0.81
LASSO	0.828	0.211	0.81
Fast Kernel SVM	0.897	-	0.92
Coxph	0.5	0.332	0.5
10-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.878	0.162	0.91
Survival Tree	0.741	0.199	0.75
Component Wise GBM	0.778	0.203	0.82
GBM	0.861	0.139	0.91
ElasticNET	0.865	0.201	0.89
LASSO	0.841	0.210	0.84
Fast Kernel SVM	0.901	-	0.95
Coxph	0.5	0.411	0.5

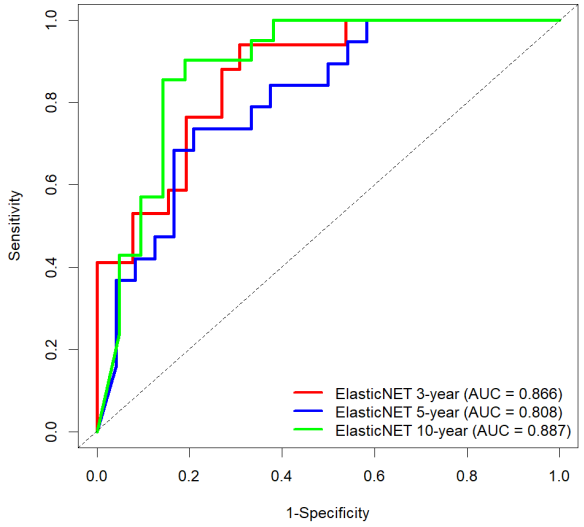


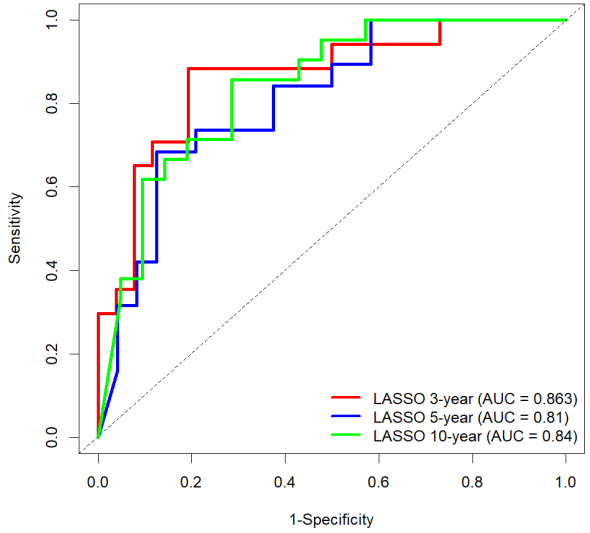


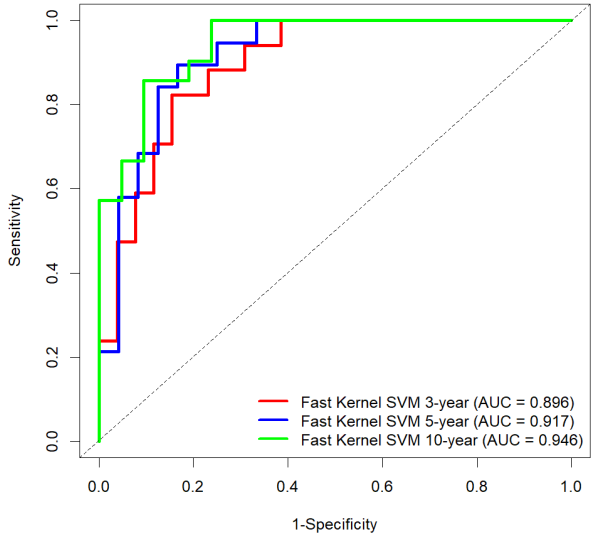


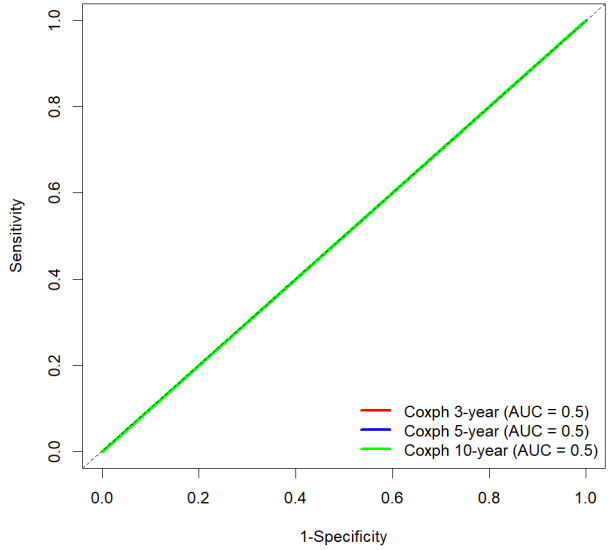
Model Performance in Male Cohort (Internal Validation)





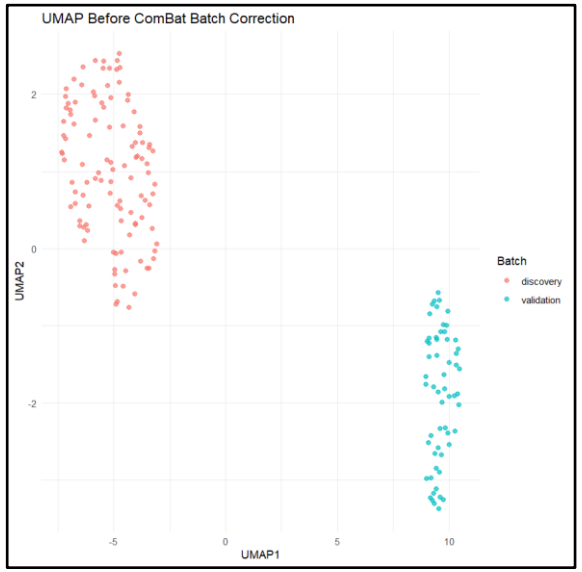


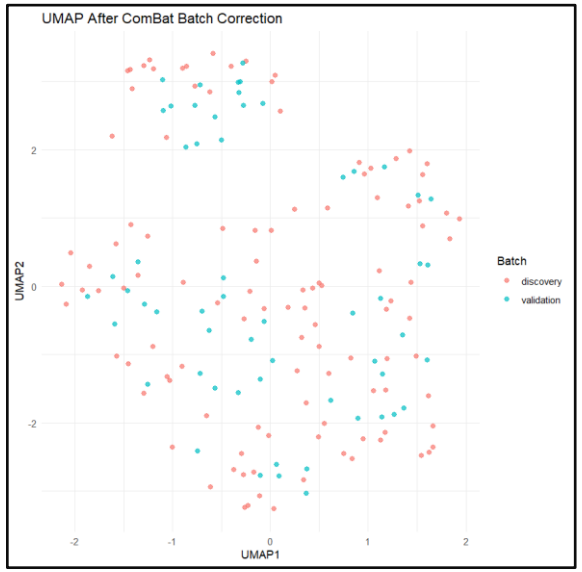


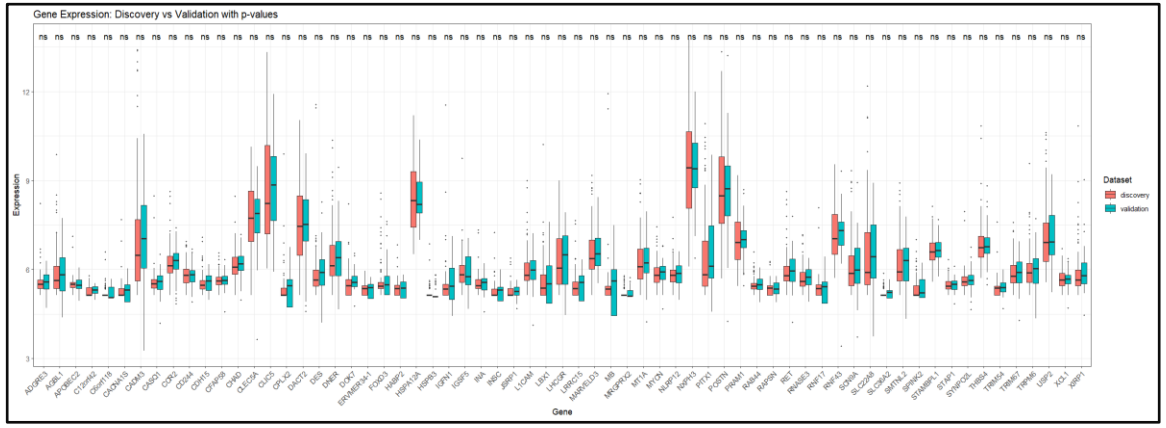


Harmonization of Data via ComBat

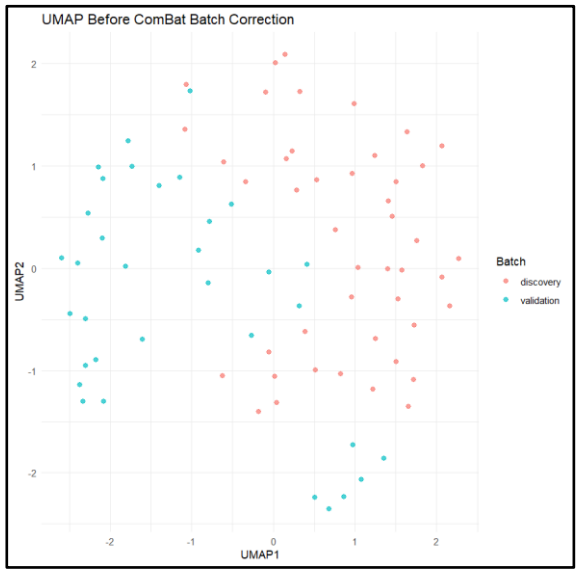
Female Cohort

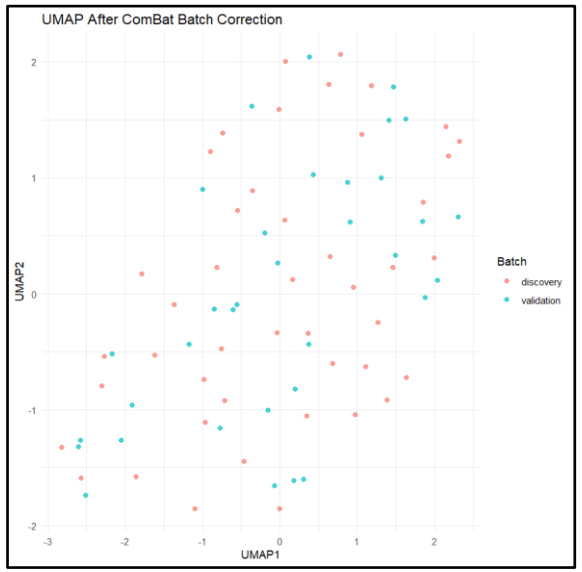


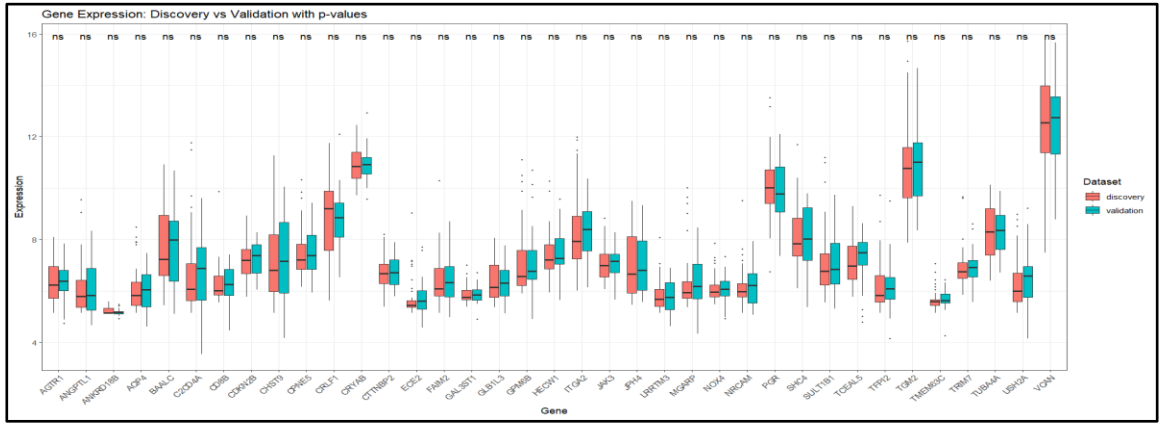




Male Cohort

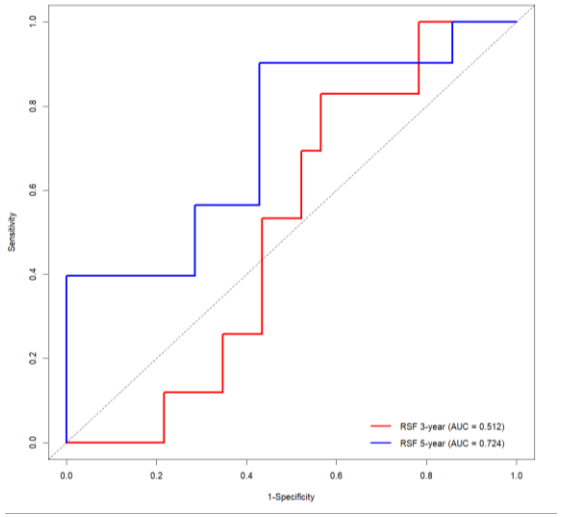


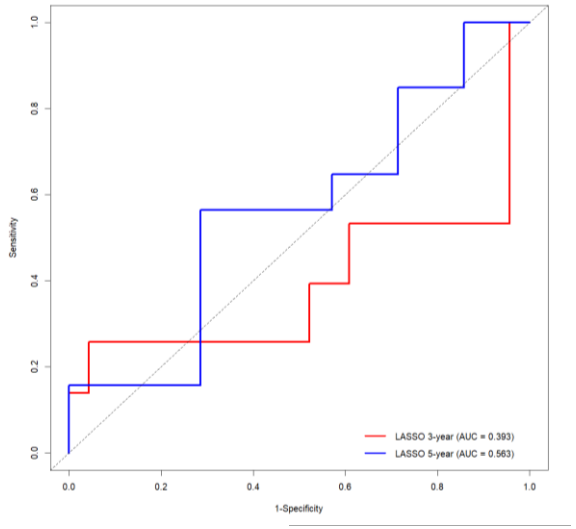


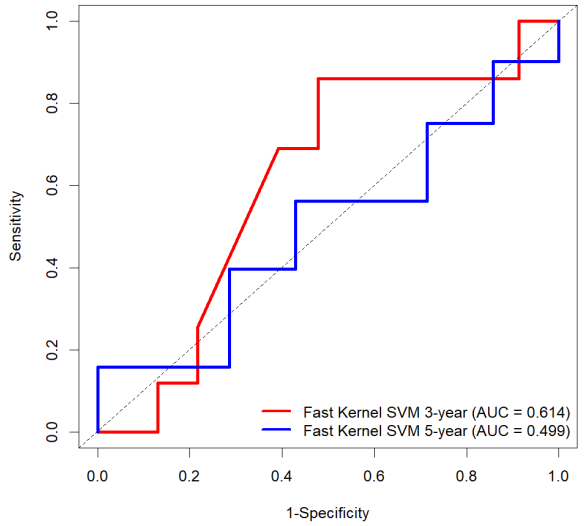


Model Performance in Female Cohort (External Validation)

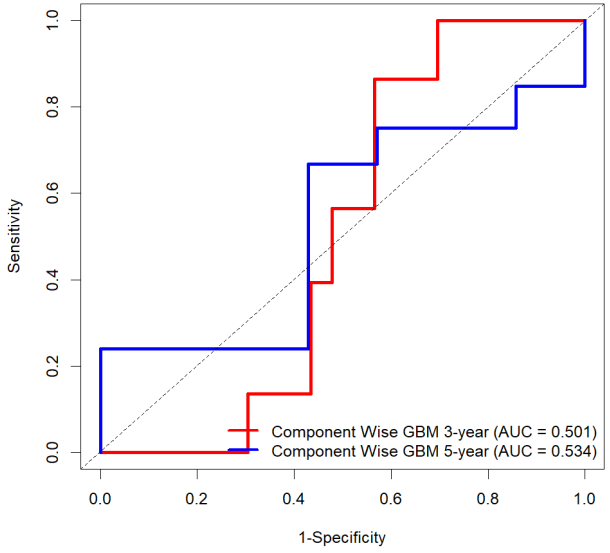
3-year			
ML model	Uno's Concordance index	Integrated Brier Score	AUC
RSF	0.544	0.052	0.51
Survival Tree	0.402	0.065	0.41
Component Wise GBM	0.456	0.048	0.50
GBM	0.607	0.048	0.51
ElasticNET	0.411	0.051	0.37
LASSO	0.444	0.061	0.39
Fast Kernel SVM	0.524	-	0.61
Coxph	0.5	0.077	0.5
5-year			
ML model	Uno's Concordance index	Integrated Brier Score	AUC
RSF	0.607	0.070	0.72
Survival Tree	0.493	0.083	0.54
Component Wise GBM	0.420	0.073	0.53
GBM	0.600	0.071	0.53
ElasticNET	0.417	0.081	0.36
LASSO	0.553	0.077	0.56
Fast Kernel SVM	0.513	-	0.50
Coxph	0.5	0.09	0.5

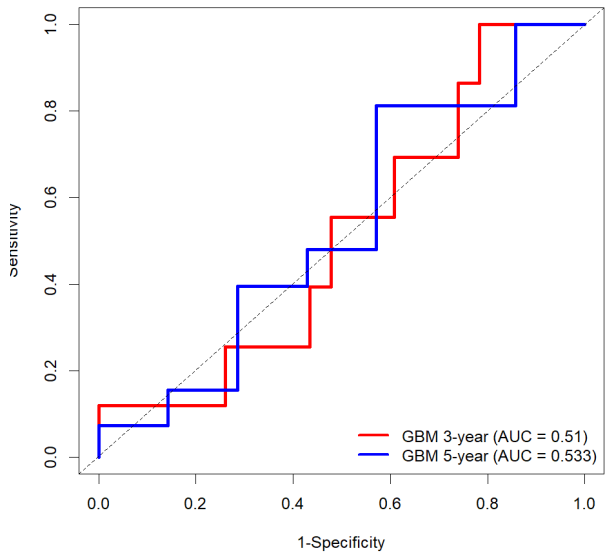


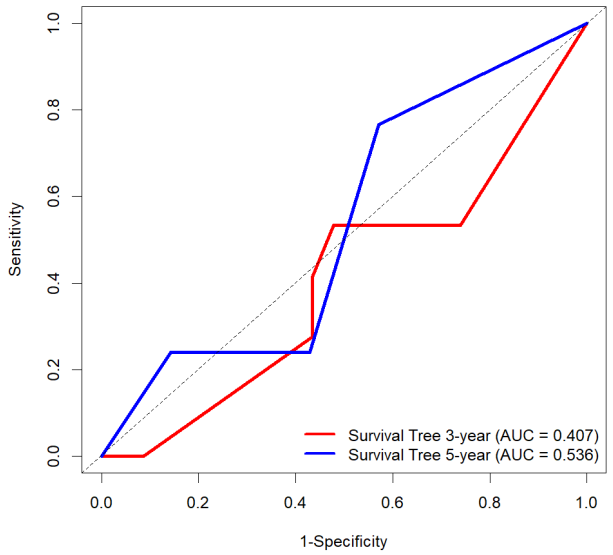


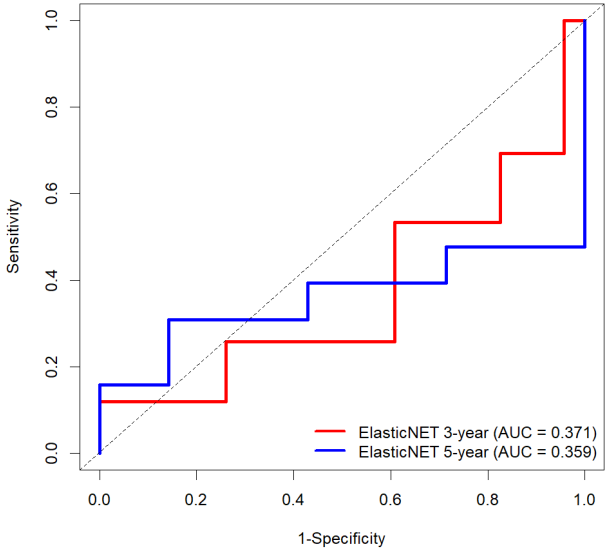


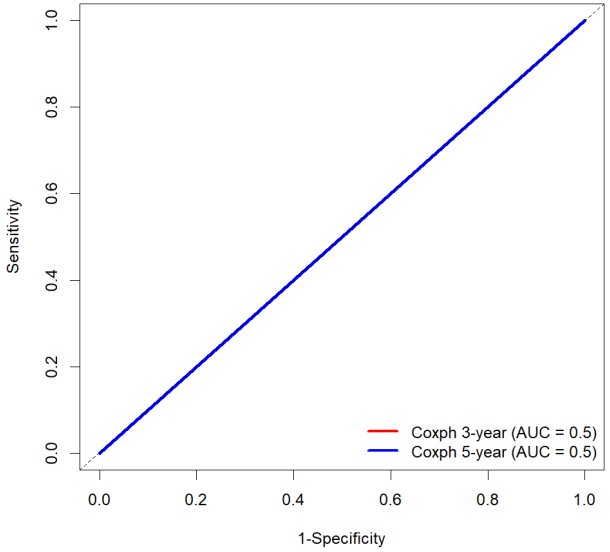
Model Performance in Female Cohort (External Validation)





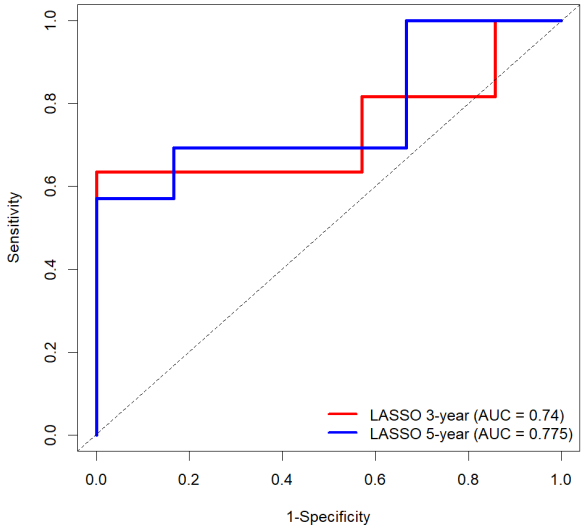


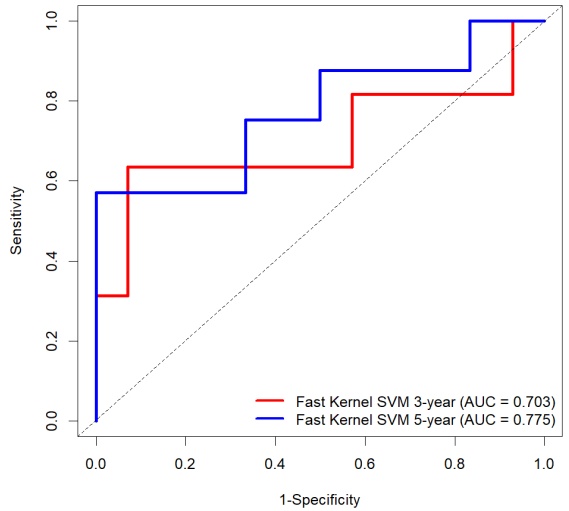


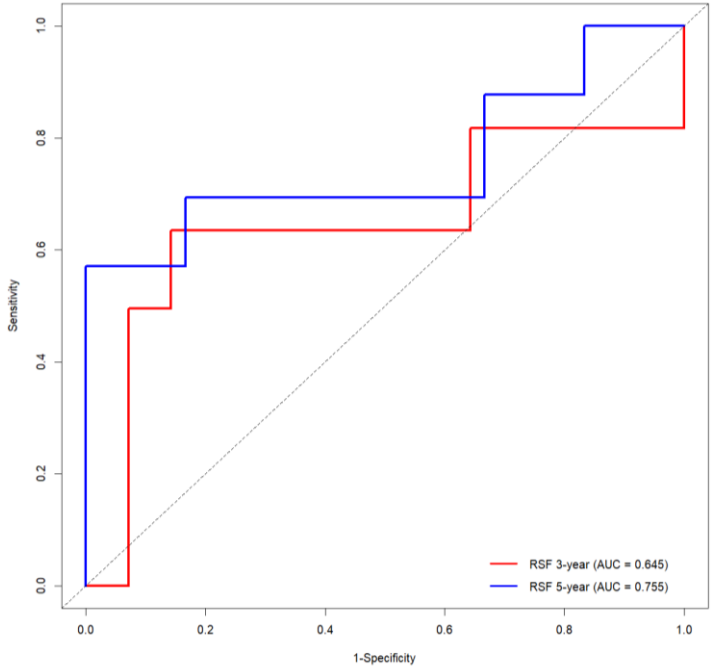


Model Performance in Male Cohort (External Validation)

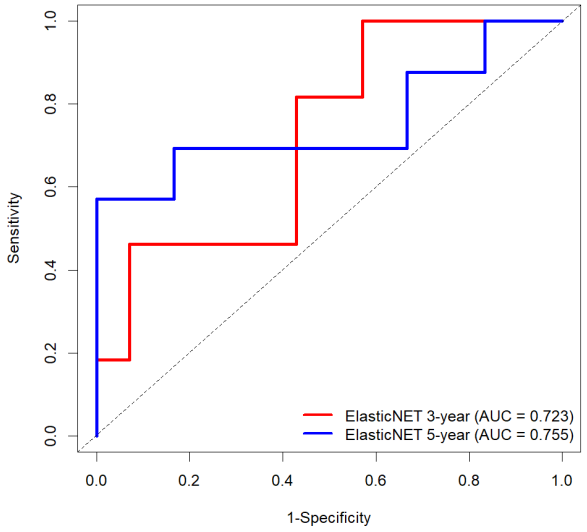
3-year			
ML model	Uno's Concordance index	Integrated Brier Score	AUC
RSF	0.653	0.071	0.65
Survival Tree	0.573	0.094	0.51
Component-wise GBM	0.653	0.068	0.59
GBM	0.677	0.075	0.62
ElasticNET	0.694	0.095	0.72
LASSO	0.798	0.054	0.74
Fast Kernel SVM	0.750	-	0.70
Coxph	0.5	0.080	0.5
5-year			
ML model	Uno's Concordance index	Integrated Brier Score	AUC
RSF	0.658	0.084	0.76
Survival Tree	0.473	0.163	0.38
Component-wise GBM	0.637	0.089	0.57
GBM	0.788	0.062	0.84
ElasticNET	0.784	0.073	0.76
LASSO	0.784	0.076	0.78
Fast Kernel SVM	0.767	-	0.78
Coxph	0.5	0.144	0.5

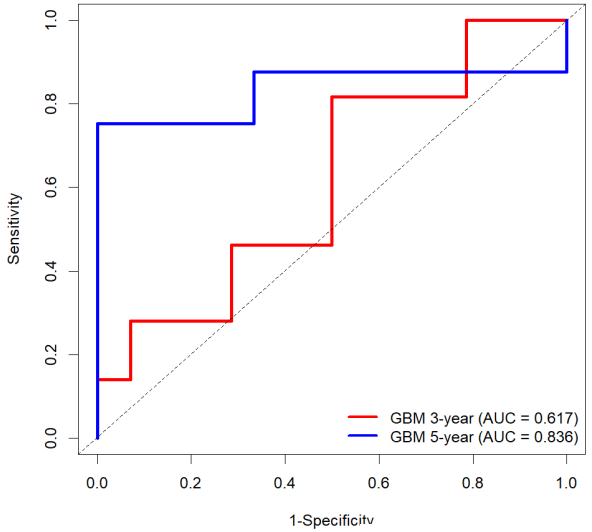


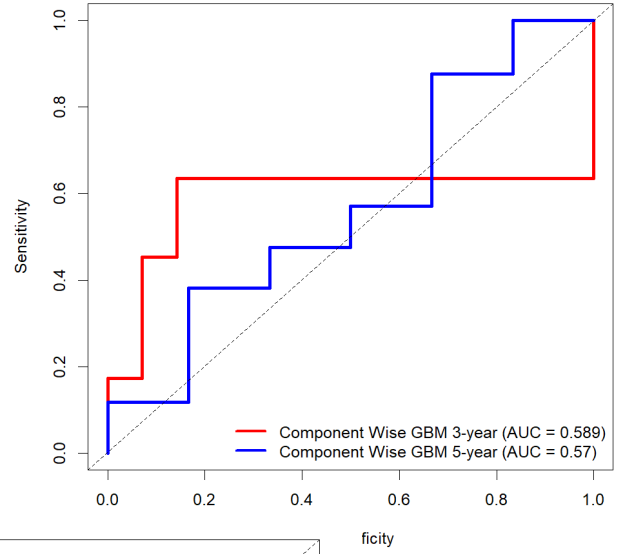


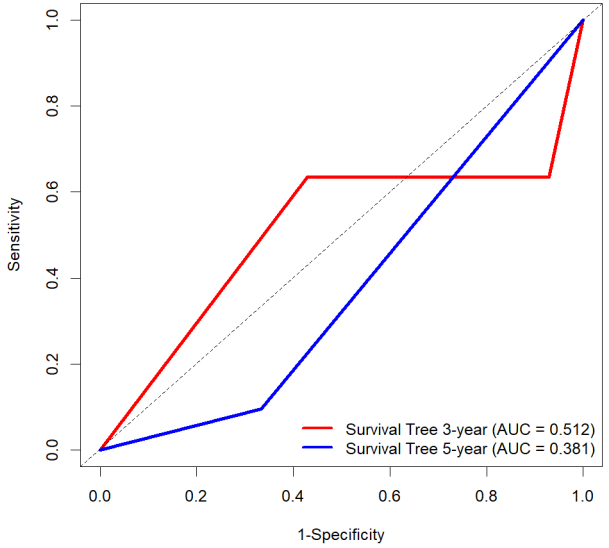


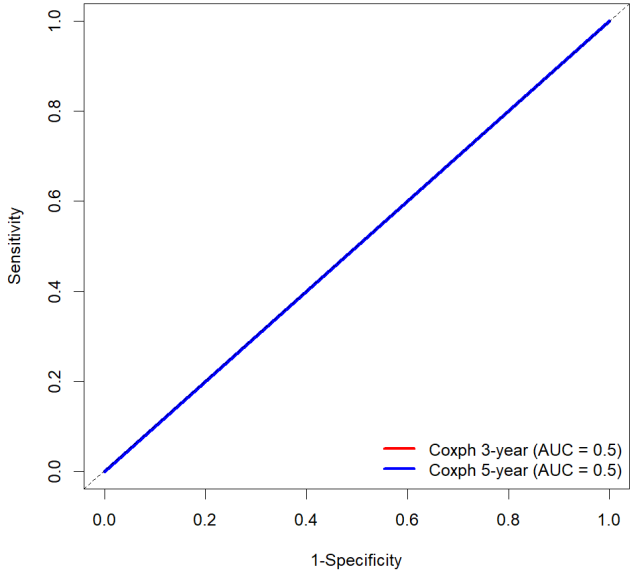
Model Performance in Male Cohort (External Validation)





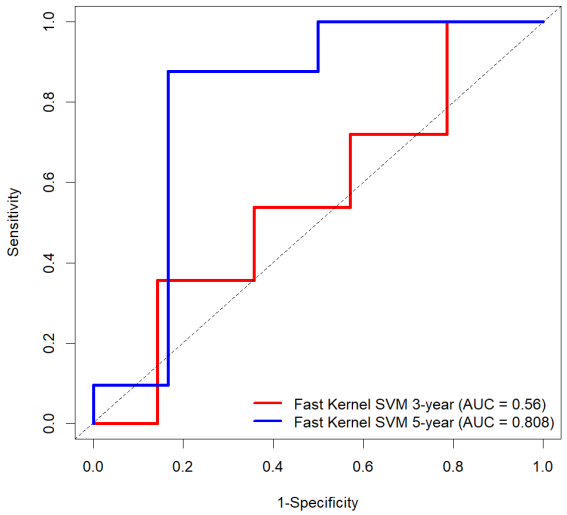


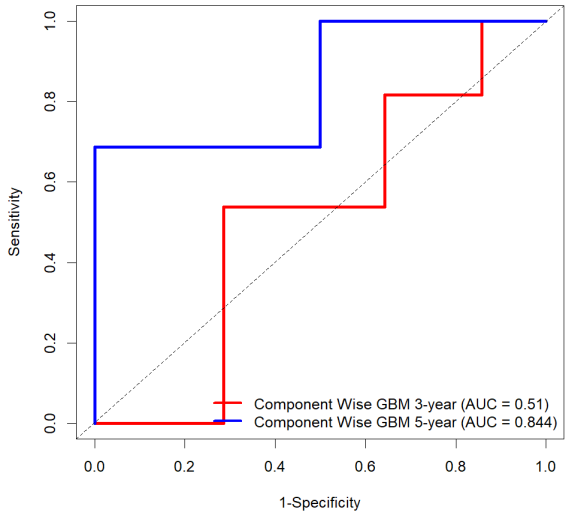


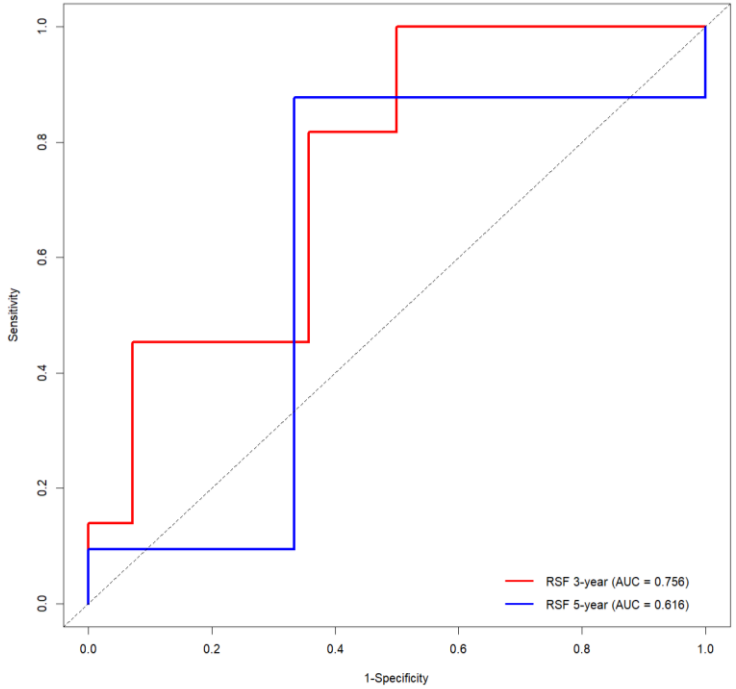


Cross-Gender Model Evaluation: Validation of Female-Specific Signatures in Male Cohort

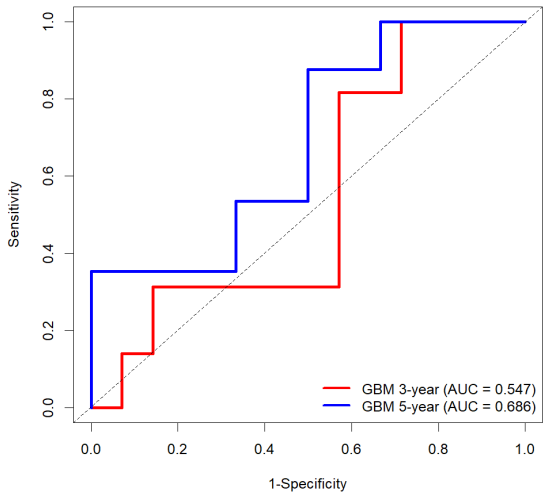
3-year			
ML model	Uno's Concordance index	Integrated Brier Score	AUC
RSF	0.774	0.604	0.76
Survival Tree	0.528	0.080	0.54
Component-wise GBM	0.484	0.066	0.51
GBM	0.573	0.065	0.55
ElasticNET	0.548	0.077	0.51
LASSO	0.516	0.067	0.46
Fast Kernel SVM	0.520	-	0.56
Coxph	0.5	0.053	0.5
5-year			
ML model	Uno's Concordance index	Integrated Brier Score	AUC
RSF	0.685	0.093	0.62
Survival Tree	0.408	0.119	0.40
Component Wise GBM	0.589	0.105	0.84
GBM	0.623	0.103	0.69
ElasticNET	0.610	0.105	0.70
LASSO	0.712	0.098	0.67
Fast Kernel SVM	0.781	-	0.81
Coxph	0.5	0.139	0.5

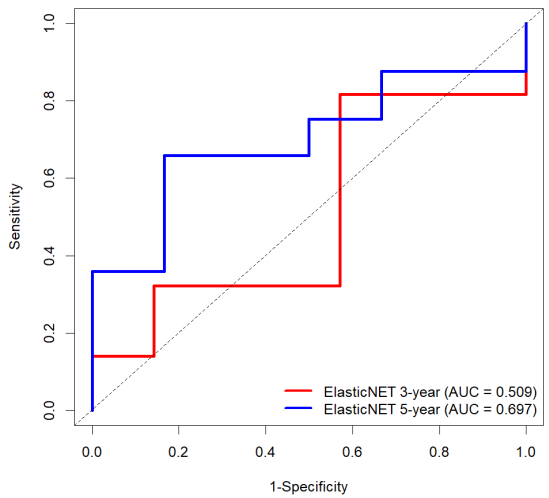


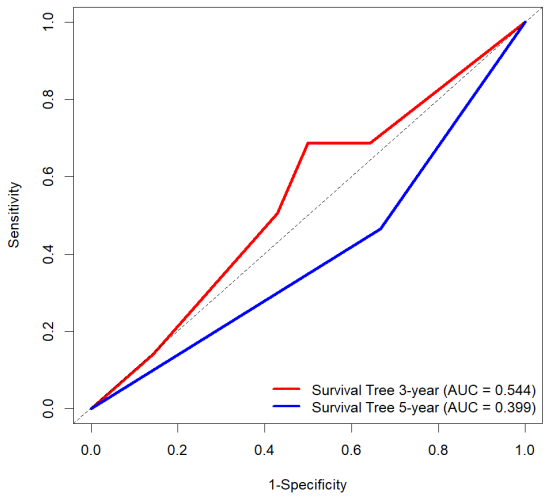


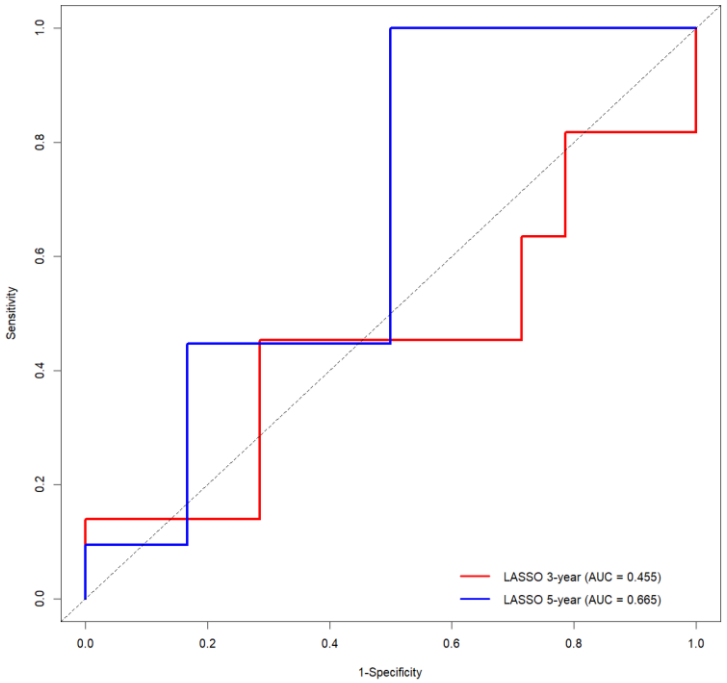


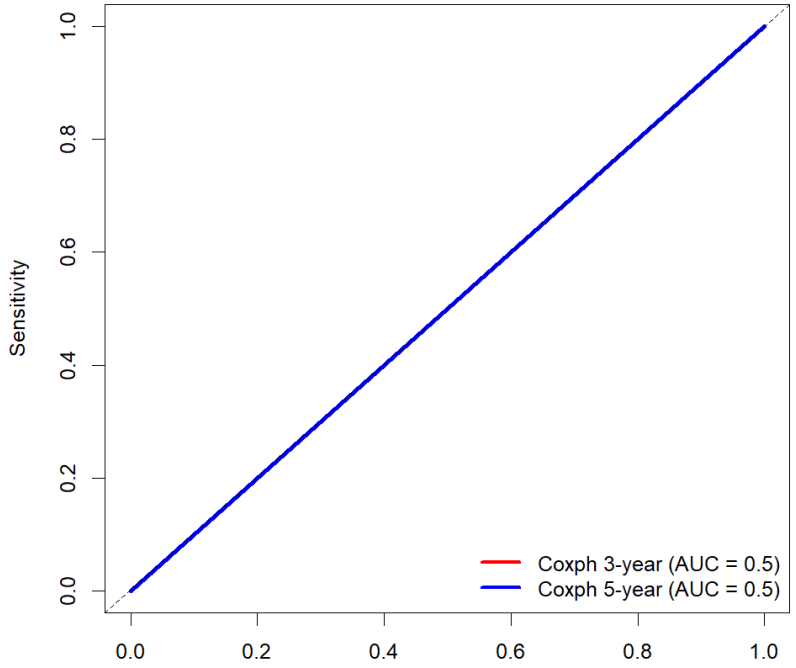
Cross-Gender Model Evaluation: Validation of Female-Specific Signatures in Male Cohort





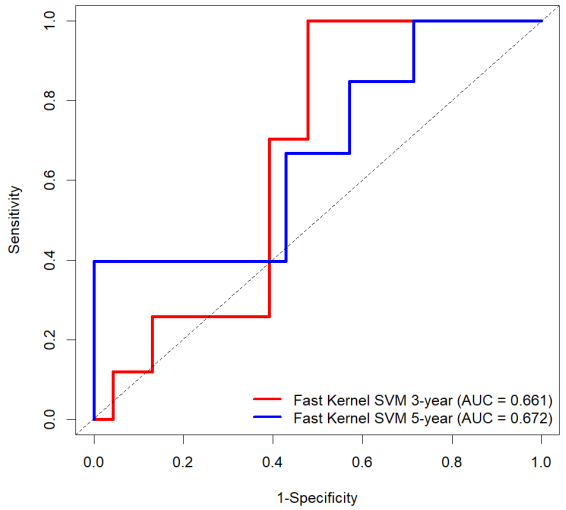


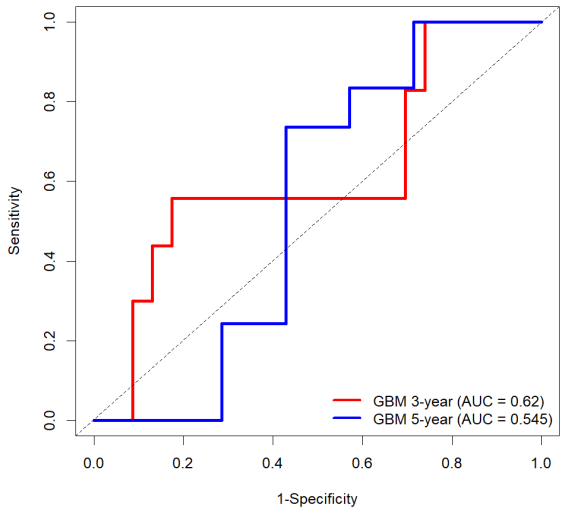


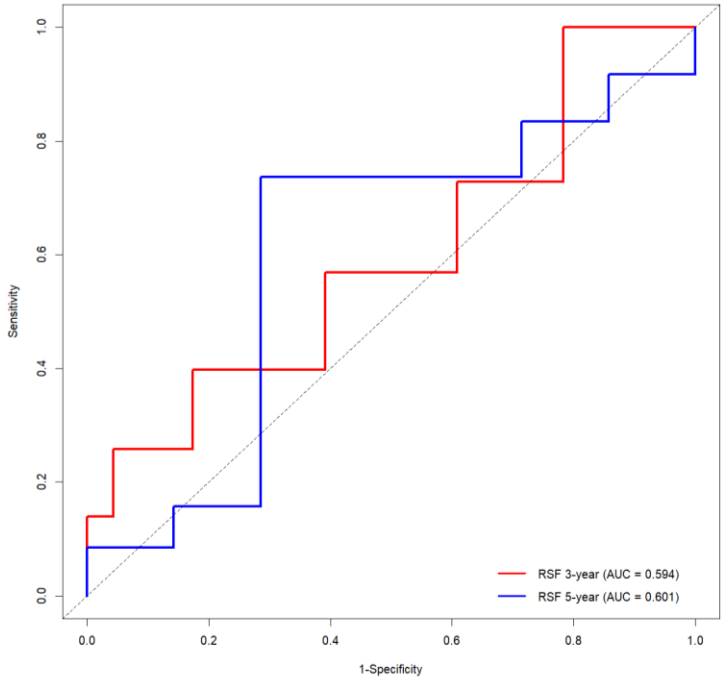


Cross-Gender Model Evaluation: Validation of Male-Specific Signatures in Female Cohort

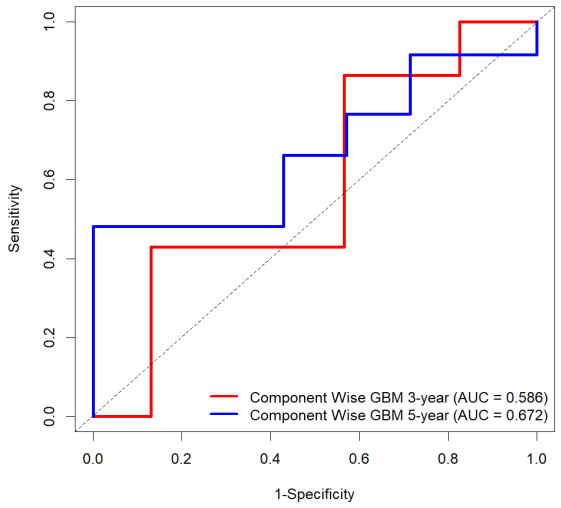
3-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.582	0.059	0.59
Survival Tree	0.487	0.083	0.44
Component Wise GBM	0.559	0.058	0.59
GBM	0.604	0.071	0.62
ElasticNET	0.457	0.140	0.53
LASSO	0.567	0.068	0.57
Fast Kernel SVM	0.622	-	0.66
Coxph	0.500	0.055	0.50
5-year			
ML model	**Uno's Concordance index**	**Integrated Brier Score**	**AUC**
RSF	0.590	0.077	0.60
Survival Tree	0.610	0.122	0.68
Component Wise GBM	0.657	0.069	0.67
GBM	0.563	0.086	0.55
ElasticNET	0.490	0.124	0.25
LASSO	0.467	0.127	0.23
Fast Kernel SVM	0.627	-	0.67
Coxph	0.500	0.094	0.50

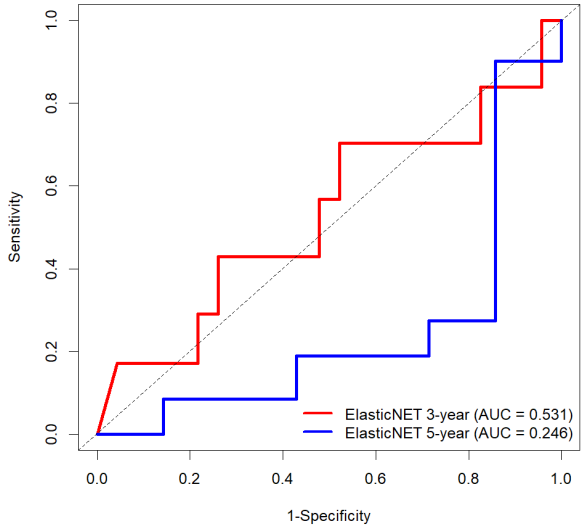


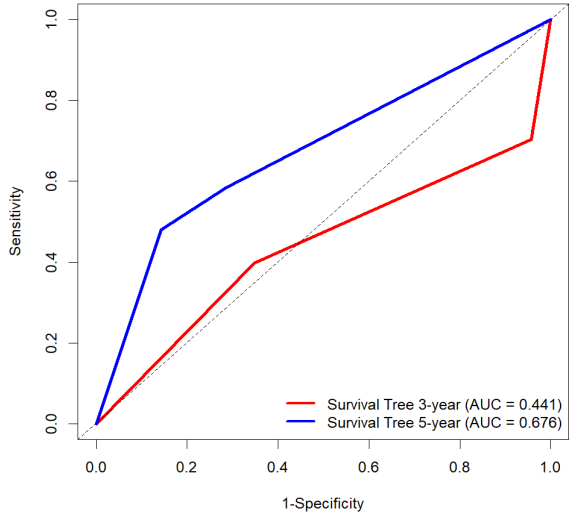


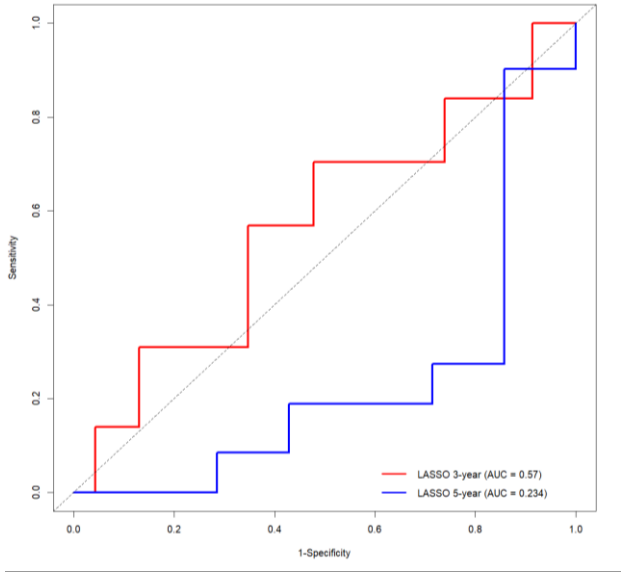


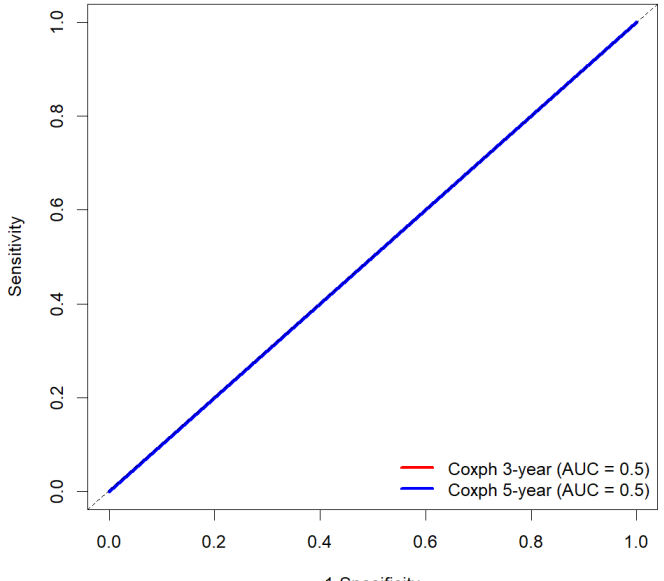
Cross-Gender Model Evaluation: Validation of Male-Specific Signatures in Female Cohort











Conclusions

Meningioma Recurrence Is Driven by Sex-Specific Molecular Mechanisms

This study reveals that male and female patients exhibit fundamentally different gene expression profiles in recurrent meningiomas. The minimal overlap in differentially expressed genes and even opposing trends for shared genes demonstrates distinct biological drivers of recurrence. This finding confirms that meningioma recurrence is not a uniform process and must be studied through a sex-stratified lens.

Prognostic Biomarkers Differ Completely Between Males and Females

No overlap in recurrence-associated genes between the sexes, in Univariate Cox regression analysis, strongly suggests that using a single, sex-agnostic prognostic model could be misleading or ineffective. Prognosis and therapeutic decisions should therefore consider sex-specific biomarkers.

Poor Cross-Sex Generalizability Highlights the Need for Sex-Specific Predictive Models

Models trained on one sex performed poorly when applied to the other. This performance loss during cross-gender external validation demonstrates the lack of model transferability across sexes. As a result, sex-specific model development is critical for clinical reliability and generalizability.

Model Selection Determines the Success of Machine Learning in Prognosis

This study demonstrates that ML models can be valuable tools for predicting RFS using transcriptomic data. However, the performance varied significantly across different ML models. This highlights the importance of careful model selection and validation to ensure reliability and clinical utility. ML offers great potential, but its application in prognosis must be approached with caution and rigorous evaluation.

Prospective Plans

- Incorporate S**ingle-cell and Spatial Transcriptomics** to capture cell-type and regional gene expression differences.
- Use **multi-omic approaches** (e.g., genomics, epigenomics, proteomics) for deeper biological insights.
- Develop **sex-specific predictive models** and improve model interpretability using tools like SHAP or LIME.
- Perform **prospective validation** in diverse clinical cohorts for real-world applicability.
- Integrate **clinical and imaging data** with transcriptomics for robust, radiogenomic prognostic tools.

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